Chapter 2

Basics of point-referenced data models

In this chapter we present the essential elements of spatial models and classical analysis for point-referenced data. As mentioned in Chapter 1, the fundamental concept underlying the theory is a stochastic process $\{Y(\mathbf{s}) : \mathbf{s} \in D\}$, where D is a fixed subset of r-dimensional Euclidean space. Note that such stochastic processes have a rich presence in the time series literature, where $r = 1$. In the spatial context, usually we encounter r to be 2 (say, northings and eastings) or 3 (e.g., northings, eastings, and altitude above sea level). For situations where $r > 1$, the process is often referred to as a *spatial process*. For example, $Y(\mathbf{s})$ may represent the level of a pollutant at site $\mathbf{s}$. While it is conceptually sensible to assume the existence of a pollutant level at all possible sites in the domain, in practice the data will be a partial realization of that spatial process. That is, it will consist of measurements at a finite set of locations, say $\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}$, where there are monitoring stations. The problem facing the statistician is inference about the spatial process $Y(\mathbf{s})$ and prediction at new locations, based upon this partial realization. The remarkable feature of the models we employ here is that, despite only seeing the process, equivalently, the spatial surface at a finite set of locations, we can infer about the surface at an uncountable number of locations. The reason is that we specify association through *structured* dependence which enables this broad interpolation.

This chapter is organized as follows. We begin with a survey of the building blocks of point-level data modeling, including stationarity, isotropy, and variograms (and their fitting via traditional moment-matching methods). We defer theoretical discussion of the spatial (typically Gaussian) process modeling that enables likelihood (and Bayesian) inference in these settings to Chapters 3 and 6. Then, we illustrate helpful exploratory data analysis tools, as well as traditional classical methods, especially kriging (point-level spatial prediction). We view all of these activities in an exploratory fashion, i.e., as a prelude to fully model-based inference under a hierarchical model. We close with some short tutorials in `R` using easy to use and widely available point-level spatial statistical analysis packages.

The material we cover in this chapter is traditionally known as *geostatistics*, and could easily fill many more pages than we devote to it here. While we prefer the more descriptive term "point-level spatial modeling," we will at times still use "geostatistics" for brevity and perhaps consistency when referencing the literature.

2.1 Elements of point-referenced modeling

2.1.1 Stationarity

For our discussion we assume that our spatial process has a mean, say $\mu(\mathbf{s}) = E(Y(\mathbf{s}))$, associated with it and that the variance of $Y(\mathbf{s})$ exists for all $\mathbf{s} \in D$. The process $Y(\mathbf{s})$ is said to be *Gaussian* if, for any $n \geq 1$ and any set of sites $\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}$, $\mathbf{Y} = (Y(\mathbf{s}_1), \ldots, Y(\mathbf{s}_n))^T$ has a multivariate normal distribution. The process is said to be *strictly stationary* (sometimes *strong* stationarity) if, for any given $n \geq 1$, any set of n sites $\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}$ and any $\mathbf{h} \in \Re^r$,

the distribution of $(Y(\mathbf{s}_1), \ldots, Y(\mathbf{s}_n))$ is the same as that of $(Y(\mathbf{s}_1 + \mathbf{h}), \ldots, Y(\mathbf{s}_n + \mathbf{h}))$. Here D is envisioned as $\Re^r$ as well.

A less restrictive condition is given by *weak stationarity* (also called second-order stationarity). A spatial process is called weakly stationary if $\mu(\mathbf{s}) \equiv \mu$ (i.e., it has a constant mean) and $\mathrm{Cov}(Y(\mathbf{s}), Y(\mathbf{s}+\mathbf{h})) = C(\mathbf{h})$ for all $\mathbf{h} \in \Re^r$ such that $\mathbf{s}$ and $\mathbf{s}+\mathbf{h}$ both lie within D. In fact, for stationarity as a second-order property we will need only the second property; $E(Y(\mathbf{s}))$ need not equal $E(Y(\mathbf{s}+\mathbf{h}))$. But since we will apply the definition only to a mean 0 spatial residual process, this distinction is not important for us. Weak stationarity implies that the covariance relationship between the values of the process at any two locations can be summarized by a covariance function $C(\mathbf{h})$, and this function depends only on the separation vector $\mathbf{h}$. Note that with all variances assumed to exist, strong stationarity implies weak stationarity. The converse is not true in general, but it *does* hold for Gaussian processes; see Exercise 4.

We offer a simple illustration of a weakly stationary process that is not strictly stationary. It is easy to see in the one-dimensional case. Suppose the process $Y_t, t = 1, 2, ...$ consists of a sequence of independent variables such that for t odd, Y_t is a binary variable taking the values 1 and -1 each with probability .5 while for t even, Y_t is normal with mean 0 and variance 1. We have weak stationarity since $\mathrm{cov}(Y(t), Y(t')) = 0, t \neq t', = 1, t = t'$. That is, we only need to know the value of $t - t'$ to specify the covariance. However, clearly Y_1, Y_3 does not have the same distribution as Y_2, Y_4.

2.1.2 Variograms

There is a third type of stationarity called *intrinsic* stationarity. Here we assume $E[Y(\mathbf{s}+\mathbf{h}) - Y(\mathbf{s})] = 0$ and define

$$E[Y(\mathbf{s}+\mathbf{h}) - Y(\mathbf{s})]^2 = Var(Y(\mathbf{s}+\mathbf{h}) - Y(\mathbf{s})) = 2\gamma(\mathbf{h}) \ . \tag{2.1}$$

Equation (2.1) makes sense only if the left-hand side depends *solely* on $\mathbf{h}$ (so that the right-hand side can be written at all), and not the particular choice of $\mathbf{s}$. If this is the case, we say the process is *intrinsically stationary*. The function $2\gamma(\mathbf{h})$ is then called the *variogram*, and $\gamma(\mathbf{h})$ is called the *semivariogram*. We can offer some intuition behind the variogram but it really arose simply as a result of its appearance in traditional kriging where one seeks the best linear unbiased predictor, as we clarify below. Behaviorally, at short distances (small $||\mathbf{h}||$), we would expect $Y(\mathbf{s}+\mathbf{h})$ and $Y(\mathbf{h})$ to be very similar, that is, $(Y(\mathbf{s}+\mathbf{h}) - Y(\mathbf{s}))^2$ to be small. As $||\mathbf{h}||$ grows larger, we expect less similarity between $Y(\mathbf{s}+\mathbf{h})$ and $Y(\mathbf{h})$, i.e., we expect $(Y(\mathbf{s}+\mathbf{h}) - Y(\mathbf{s}))^2$ to be larger. So, a plot of $\gamma(\mathbf{h})$ would be expected to increase with $||\mathbf{h}||$, providing some insight into spatial behavior. (The covariance function $C(\mathbf{h})$ is sometimes referred to as the *covariogram*, especially when plotted graphically.) Note that intrinsic stationarity defines only the first and second moments of the differences $Y(\mathbf{s}+\mathbf{h}) - Y(\mathbf{s})$. It says nothing about the joint distribution of a collection of variables $Y(\mathbf{s}_1), \ldots, Y(\mathbf{s}_n)$, and thus provides no likelihood.

In fact, it says nothing about the moments of the $Y(\mathbf{s})$'s, much less their distribution. It only describes the behavior of differences rather than the behavior of the data that we observe, clearly unsatisfying from the perspective of data analysis. In fact, the $Y(\mathbf{s})$'s need not have any moments. For example, we might have $Y(\mathbf{s}) = W(\mathbf{s}) + V$ where $W(\mathbf{s})$ is a collection of i.i.d. normal variables and, independently, V is a Cauchy random variable. Then $Y(\mathbf{s})$ is intrinsically stationary but the $Y(\mathbf{s})$'s have no moments. Even more disconcerting, the distribution of $Y(\mathbf{s}) - Y(\mathbf{s}')$ may be proper while the distribution of $Y(\mathbf{s})$ and of $Y(\mathbf{s}')$ may be improper. For instance, suppose the joint distribution, $f(Y(\mathbf{s}), Y(\mathbf{s}')) \propto e^{-(Y(\mathbf{s})-Y(\mathbf{s}'))^2/2}$. Then, $Y(\mathbf{s}) - Y(\mathbf{s}') \sim N(0,1)$ and, in fact, $f(Y(\mathbf{s})|Y(\mathbf{s}'))$ and $f(Y(\mathbf{s}')|Y(\mathbf{s}))$ are proper, $N(0,1)$ but the joint distribution is improper. How could we employ a probability specification that could not possibly have generated the data we observe?

It is easy to see the relationship between the variogram and the covariance function:

$$\begin{aligned} 2\gamma(\mathbf{h}) &= Var\left(Y\left(\mathbf{s}+\mathbf{h}\right)-Y\left(\mathbf{s}\right)\right) \\ &= Var(Y(\mathbf{s}+\mathbf{h}))+Var(Y(\mathbf{s}))-2Cov(Y(\mathbf{s}+\mathbf{h}),Y(\mathbf{s})) \\ &= C(\mathbf{0})+C(\mathbf{0})-2C\left(\mathbf{h}\right) \\ &= 2\left[C\left(\mathbf{0}\right)-C\left(\mathbf{h}\right)\right] . \end{aligned}$$

Thus,

$$\gamma\left(\mathbf{h}\right)=C\left(\mathbf{0}\right)-C\left(\mathbf{h}\right) . \tag{2.2}$$

From (2.2) we see that given C, we are able to recover γ easily. But what about the converse; in general, can we recover C from γ? Here it turns out we need to assume a bit more: if the spatial process is *ergodic*, then $C\left(\mathbf{h}\right)\rightarrow 0$ as $||\mathbf{h}||\rightarrow\infty$, where $||\mathbf{h}||$ denotes the length of the $\mathbf{h}$ vector. This is an intuitively sensible condition, since it means that the covariance between the values at two points vanishes as the points become further separated in space. But taking the limit of both sides of (2.2) as $||\mathbf{h}||\rightarrow\infty$, we then have that $lim_{||\mathbf{h}||\rightarrow\infty}\gamma\left(\mathbf{h}\right)=C\left(\mathbf{0}\right)$. Thus, using the dummy variable $\mathbf{u}$ to avoid confusion, we have

$$C\left(\mathbf{h}\right)=C(\mathbf{0})-\gamma(\mathbf{h})=lim_{||\mathbf{u}||\rightarrow\infty}\gamma\left(\mathbf{u}\right)\ -\ \gamma\left(\mathbf{h}\right) . \tag{2.3}$$

In general, the limit on the right-hand side need not exist, but if it does, then the process is weakly (second-order) stationary with $C\left(\mathbf{h}\right)$ as given in (2.3). We therefore have a way to determine the covariance function C from the semivariogram γ. Thus weak stationarity implies intrinsic stationarity, but the converse is not true; indeed, the next section offers examples of processes that are intrinsically stationary but not weakly stationary.

A valid variogram necessarily satisfies a negative definiteness condition. In fact, for any set of locations $\mathbf{s}_1,\ldots,\mathbf{s}_n$ and any set of constants $a_1,\ldots,a_n$ such that $\sum_i a_i=0$, if $\gamma(\mathbf{h})$ is valid, then

$$\sum_i\sum_j a_i a_j\gamma(\mathbf{s}_i-\mathbf{s}_j)\leq 0 . \tag{2.4}$$

To see this, note that

$$\begin{aligned} \sum_i\sum_j a_i a_j\gamma(\mathbf{s}_i-\mathbf{s}_j) &= \frac{1}{2}E\sum_i\sum_j a_i a_j(Y(\mathbf{s}_i)-Y(\mathbf{s}_j))^2 \\ &= -E\sum_i\sum_j a_i a_j Y(\mathbf{s}_i)Y(\mathbf{s}_j) \\ &= -E\left[\sum_i a_i Y(\mathbf{s}_i)\right]^2\ \leq\ 0 . \end{aligned}$$

We remark that, despite the suggestion of expression (2.2), there is no relationship between this result and the positive definiteness condition for covariance functions (see Subsection 3.1.2). Cressie (1993) discusses further necessary conditions for a valid variogram. Lastly, the condition (2.4) emerges naturally in ordinary kriging (see Section 2.4).

2.1.3 Isotropy

Another important related concept is that of isotropy. If the semivariogram function $\gamma\left(\mathbf{h}\right)$ depends upon the separation vector only through its length $||\mathbf{h}||$, then we say that the variogram is *isotropic*; that is, if $\gamma\left(\mathbf{h}\right)$ is a real-valued function of a univariate argument, and can be written as $\gamma\left(||\mathbf{h}||\right)$. If not, we say it is *anisotropic*. Because of the foregoing issues with intrinsic stationarity, in the absence of a full probabilistic specification (as detailed in

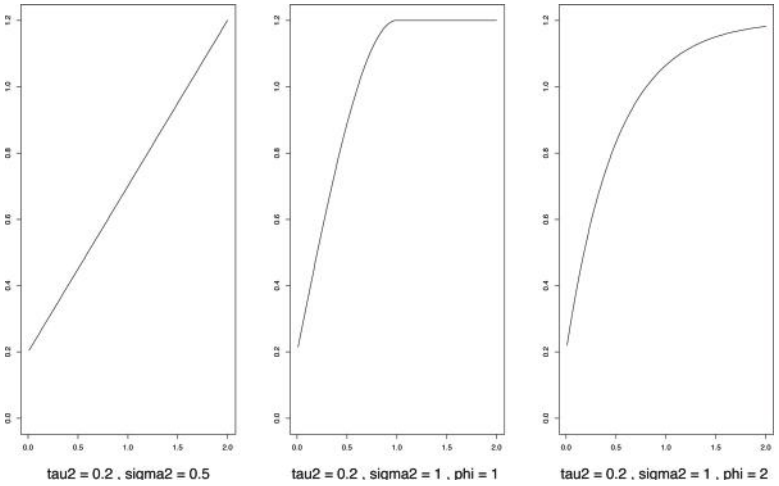


Figure 2.1 *Theoretical semivariograms for three models: (a) linear, (b) spherical, and (c) exponential.*

Chapter 3), we are reluctant to associate an isotropic variogram with a stochastic process. Nonetheless, in the literature, we find terminology stating that, if a process is intrinsically stationary and isotropic, it is also called *homogeneous*.

Isotropic variograms are popular because of their simplicity, interpretability, and, in particular, because a number of relatively simple parametric forms are available as candidates for the semivariogram. Denoting $||\mathbf{h}||$ by d for notational simplicity, we now consider a few of the more important such forms.

1. *Linear:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 d & \text{if } d > 0,\ \tau^2 > 0,\ \sigma^2 > 0 \\ 0 & \text{otherwise} \end{cases} .$$

Note that $\gamma(d) \to \infty$ as $d \to \infty$, and so this semivariogram does not correspond to a weakly stationary process (although it is intrinsically stationary). This semivariogram is plotted in Figure 2.1(a) using the parameter values $\tau^2 = 0.2$ and $\sigma^2 = 0.5$.

2. *Spherical:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 & \text{if } d \geq 1/\phi, \\ \tau^2 + \sigma^2 \left\{ \frac{3\phi d}{2} - \frac{1}{2}(\phi d)^3 \right\} & \text{if } 0 < d \leq 1/\phi, \\ 0 & \text{otherwise} \end{cases} .$$

The spherical semivariogram is valid in $r = 1, 2$, or 3 dimensions, but for $r \geq 4$ it fails to correspond to a spatial variance matrix that is positive definite (as required to specify a valid joint probability distribution). This variogram owes its popularity largely to the fact that it offers clear illustrations of the *nugget*, *sill*, and *range*, three characteristics traditionally associated with variograms. Specifically, consider Figure 2.1(b), which plots the spherical semivariogram using the parameter values $\tau^2 = 0.2$, $\sigma^2 = 1$, and $\phi = 1$. While $\gamma(0) = 0$ by definition, $\gamma(0^+) \equiv lim_{d \to 0^+} \gamma(d) = \tau^2$; this quantity is the *nugget*. Next, $\lim_{d \to \infty} \gamma(d) = \tau^2 + \sigma^2$; this asymptotic value of the semivariogram is called the *sill*. (The sill minus the nugget, which is simply σ^2 in this case, is called the *partial sill*.) Finally, the value $d = 1/\phi$ at which $\gamma(d)$ first reaches its ultimate level (the sill) is called the *range*. It is for this reason that many of the variogram models of this subsection are often parametrized through $R \equiv 1/\phi$. Confusingly, both R and ϕ are sometimes referred to as the *range* parameter, although ϕ is often more accurately referred to as the *decay* parameter.

Note that for the linear semivariogram, the nugget is τ^2 but the sill and range are both infinite. For other variograms (such as the next one we consider), the sill is finite, but only reached asymptotically.

3. *Exponential:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 \left(1 - \exp\left(-\phi d\right)\right) & \text{if } d > 0, \\ 0 & \text{otherwise} \end{cases} .$$

The exponential has an advantage over the spherical in that it is simpler in functional form while still being a valid variogram in all dimensions (and without the spherical's finite range requirement). However, note from Figure 2.1(c), which plots this semivariogram assuming $\tau^2 = 0.2$, $\sigma^2 = 1$, and $\phi = 2$, that the sill is only reached asymptotically; strictly speaking, the range $R = 1/\phi$ is infinite. In cases like this, the notion of an *effective range* is often used, i.e., the distance at which there is essentially no lingering spatial correlation. To make this notion precise, we must convert from γ scale to C scale (possible here since $\lim_{d\to\infty} \gamma(d)$ exists; the exponential is not only intrinsically but also weakly stationary). From (2.3) we have

$$\begin{aligned} C(d) &= lim_{u\to\infty}\gamma(u) - \gamma(d) \\ &= \tau^2 + \sigma^2 - \left[\tau^2 + \sigma^2(1 - \exp(-\phi d))\right] \\ &= \sigma^2 \exp(-\phi d) \ . \end{aligned}$$

Hence

$$C(t) = \begin{cases} \tau^2 + \sigma^2 & \text{if } d = 0 \\ \sigma^2 \exp(-\phi d) & \text{if } d > 0 \end{cases} . \tag{2.5}$$

If the nugget $\tau^2 = 0$, then this expression reveals that the correlation between two points d units apart is $\exp(-\phi d)$; note that $\exp(-\phi d) = 1^-$ for $d = 0^+$ and $\exp(-\phi d) = 0$ for $d = \infty$, both in concert with this interpretation.

A common definition of the *effective range*, d_0, is the distance at which this correlation is *negligible*, customarily taken as having dropped to only 0.05. Setting $\exp(-\phi d_0)$ equal to this value we obtain $t_0 \approx 3/\phi$, since $\log(0.05) \approx -3$. The range will be discussed in more detail in Subsection 3.1.2.

Finally, the form of (2.5) gives a clear example of why the nugget (τ^2 in this case) is often viewed as a "nonspatial effect variance," and the partial sill (σ^2) is viewed as a "spatial effect variance." That is, we have two variance components. Along with ϕ, a statistician would likely view fitting this model to a spatial data set as an exercise in estimating these three parameters. We shall return to variogram model fitting in Subsection 2.1.4.

4. *Gaussian:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 \left(1 - \exp\left(-\phi^2 d^2\right)\right) & \text{if } d > 0 \\ 0 & \text{otherwise} \end{cases} . \tag{2.6}$$

The Gaussian variogram is an analytic function and yields very smooth realizations of the spatial process. We shall say much more about process smoothness in Subsection 3.1.4.

5. *Powered exponential:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 \left(1 - \exp\left(-\left|\phi d\right|^p\right)\right) & \text{if } d > 0 \\ 0 & \text{otherwise} \end{cases} . \tag{2.7}$$

Here $0 < p \leq 2$ yields a family of valid variograms. Note that both the Gaussian and the exponential forms are special cases.

6. *Rational quadratic:*

$$\gamma(d) = \begin{cases} \tau^2 + \frac{\sigma^2 d^2}{(\phi + d^2)} & \text{if } d > 0 \\ 0 & \text{otherwise} \end{cases} .$$

Model	Covariance function, $C(d)$
Linear	$C(d)$ does not exist
Spherical	$C(d) = \begin{cases} 0 & \text{if } d \geq 1/\phi \\ \sigma^2\left[1 - \frac{3}{2}\phi d + \frac{1}{2}(\phi d)^3\right] & \text{if } 0 < d \leq 1/\phi \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Exponential	$C(d) = \begin{cases} \sigma^2 \exp(-\phi d) & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Powered exponential	$C(d) = \begin{cases} \sigma^2 \exp(-\lvert\phi d\rvert^p) & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Gaussian	$C(d) = \begin{cases} \sigma^2 \exp(-\phi^2 d^2) & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Rational quadratic	$C(d) = \begin{cases} \sigma^2\left(1 - \frac{d^2}{(\phi + d^2)}\right) & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Wave	$C(d) = \begin{cases} \sigma^2 \frac{\sin(\phi d)}{\phi d} & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Power law	$C(d)$ does not exist
Matérn	$C(d) = \begin{cases} \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (\phi d)^\nu K_\nu(\phi d) & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$
Matérn at $\nu = 3/2$	$C(d) = \begin{cases} \sigma^2 (1 + \phi d) \exp(-\phi d) & \text{if } d > 0 \\ \tau^2 + \sigma^2 & d = 0 \end{cases}$

Table 2.1 *Summary of common isotropic parametric covariance functions (covariograms).*

7. *Wave:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2\left(1 - \frac{\sin(\phi d)}{\phi d}\right) & \text{if } d > 0 \\ 0 & \text{otherwise} \end{cases}.$$

Note this is an example of a variogram that is not monotonically increasing. The associated covariance function is $C(d) = \sigma^2 \sin(\phi d)/(\phi d)$. Bessel functions of the first kind include the wave covariance function and are discussed in detail in Subsections 3.1.2 and 6.1.3.

8. *Power law:*

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 d^\lambda & \text{of } d > 0 \\ 0 & d = 0 \end{cases}.$$

This generalizes the linear case and produces valid intrinsic (albeit not weakly) stationary semivariograms provided $0 \leq \lambda < 2$.

9. *Matérn:* The variogram for the Matérn class is given by

$$\gamma(d) = \begin{cases} \tau^2 + \sigma^2\left[1 - \frac{(2\sqrt{\nu}d\phi)^\nu}{2^{\nu-1}\Gamma(\nu)} K_\nu(2\sqrt{\nu}d\phi)\right] & \text{if } d > 0 \\ 0 & d = 0 \end{cases}. \tag{2.8}$$

This class was originally suggested by Matérn (1960, 1986). Interest in it was revived by Handcock and Stein (1993) and Handcock and Wallis (1994), who demonstrated attractive interpretations for ν as well as ϕ. In particular, $\nu > 0$ is a parameter controlling the smoothness of the realized random field (see Subsection 3.1.4) while ϕ is a spatial decay parameter. The function $\Gamma(\cdot)$ is the usual gamma function while K_ν is the modified Bessel function of order ν (see, e.g., Abramowitz and Stegun, 1965, Chapter 9). Implementations of this function are available in several `C/C++` libraries and also in `R` packages such as `spBayes` and `geoR`. Note that special cases of the above are the exponential ($\nu = 1/2$) and the Gaussian ($\nu \to \infty$). At $\nu = 3/2$ we obtain a closed form as well, namely $\gamma(d) =$

Model	Variogram, $\gamma(d)$
Linear	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 d & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Spherical	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 & \text{if } d \geq 1/\phi \\ \tau^2 + \sigma^2 \left[\frac{3}{2}\phi d - \frac{1}{2}(\phi d)^3\right] & \text{if } 0 < d \leq 1/\phi \\ 0 & d = 0 \end{cases}$
Exponential	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2(1 - \exp(-\phi d)) & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Powered exponential	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2(1 - \exp(-\lvert\phi d\rvert^p)) & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Gaussian	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2(1 - \exp(-\phi^2 d^2)) & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Rational quadratic	$\gamma(d) = \begin{cases} \tau^2 + \frac{\sigma^2 d^2}{(\phi + d^2)} & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Wave	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2(1 - \frac{\sin(\phi d)}{\phi d}) & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Power law	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 d^\lambda & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Matérn	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 \left[1 - \frac{(\phi d)^\nu}{2^{\nu-1}\Gamma(\nu)} K_\nu(\phi d)\right] & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$
Matérn at $\nu = 3/2$	$\gamma(d) = \begin{cases} \tau^2 + \sigma^2 \left[1 - (1 + \phi d)\exp(-\phi d)\right] & \text{if } d > 0 \\ 0 & d = 0 \end{cases}$

Table 2.2 *Summary of common parametric isotropic variograms.*

$\tau^2 + \sigma^2 [1 - (1 + \phi d) \exp(-\phi d)]$ for $t > 0$. In fact, we obtain a polynomial times exponential from for the Matérn for all ν of the form, $\nu = k + \frac{1}{2}$ with k a non-negative integer. The Matérn covariance function is often reparametrized to $\alpha = 2\sqrt{\nu}\phi$ along with $\eta = \sigma^2\phi^{2\nu}$ and ν. This transformation is helpful in providing better behaved model fitting, particularly using Markov chain Monte Carlo (see Chapter 6 for further discussion).

The covariance functions and variograms we have described in this subsection are conveniently summarized in Tables 2.1 and 2.2, respectively. An important point which the reader may wonder about is the fact that every presented covariance function and variogram has a discontinuity at 0. That is, the limit as $d \to 0$ for the covariance function is σ^2 not $\sigma^2 + \tau^2$ and for the variogram is τ^2, not 0. Evidently, this is not a typo! What is the reason? We elaborate this in greater detail in Chapter 3. Here, we offer a simple explanation. Consider the form of the residual for a spatial model. We might write it as $r(\mathbf{s}) = w(\mathbf{s}) + \epsilon(\mathbf{s})$ where $w(\mathbf{s})$ is a spatial process, say a stationary Gaussian process with mean 0 and covariance function $\sigma^2\rho(\mathbf{s} - \mathbf{s}')$ and $\epsilon(\mathbf{s})$ is a pure error process, i.e., the $\epsilon(\mathbf{s})$'s are i.i.d., say $N(0, \tau^2)$ with the ϵ's independent of the w's. Then, it is straightforward to compute $\text{cov}(r(\mathbf{s}), r(\mathbf{s}')) = \sigma^2\rho(\mathbf{s} - \mathbf{s}')$ while $\text{var}Y(\mathbf{s}) = \sigma^2 + \tau^2$, whence the discontinuity at 0 emerges. And, in fact, $\text{corr}(r(\mathbf{s}), r(\mathbf{s}'))$ is bounded by $\sigma^2/(\sigma^2 + \tau^2)$. The rationale for including the $\epsilon(\mathbf{s})$'s into the model is that we don't want to insist that all model error is spatial. Of course, we certainly want to include the w's in order to be able to capture a spatial story. Possible explanations for the pure error contribution include: (i) measurement error associated with the data collection at a given location, (ii) replication error to express the possibility that repeated measurements at the same location might not provide identical observations, or (iii) micro-scale error to acknowledge that, though we never see

observations closer to each other than the minimum pairwise distance in our sample, there might be very fine scale structure represented as noise.

2.1.4 Variogram model fitting

Having seen a fairly large selection of models for the variogram, one might well wonder how we choose one of them for a given data set, or whether the data can really distinguish them (see Subsection 6.1.3 in this latter regard). Historically, a variogram model is chosen by plotting the *empirical semivariogram* (Matheron, 1963), a simple nonparametric estimate of the semivariogram, and then comparing it to the various theoretical shapes available from the choices in the previous subsection. The customary empirical semivariogram is

$$\widehat{\gamma}(d) = \frac{1}{2N(d)} \sum_{(\mathbf{s}_i,\mathbf{s}_j)\in N(d)} [Y(\mathbf{s}_i) - Y(\mathbf{s}_j)]^2 \ , \tag{2.9}$$

where $N(d)$ is the set of pairs of points such that $||\mathbf{s}_i - \mathbf{s}_j|| = d$, and $|N(d)|$ is the number of pairs in this set. Notice that, unless the observations fall on a regular grid, the distances between the pairs will all be different, so this will not be a useful estimate as it stands. Instead we would "grid up" the d-space into intervals $I_1 = (0, d_1), I_2 = (d_1, d_2)$, and so forth, up to $I_K = (d_{K-1}, d_K)$ for some (typically regular) grid $0 < d_1 < \cdots < d_K$. Representing the d values in each interval by the midpoint of the interval, we then alter our definition of $N(d)$ to

$$N(d_k) = \{(\mathbf{s}_i, \mathbf{s}_j) : ||\mathbf{s}_i - \mathbf{s}_j|| \in I_k\} \ , \ k = 1, \ldots, K \ .$$

Selection of an appropriate number of intervals K and location of the upper endpoint t_K is reminiscent of similar issues in histogram construction. Journel and Huijbregts (1979) recommend bins wide enough to capture at least 30 pairs per bin.

Clearly (2.9) is nothing but a method of moments (MOM) estimate, the semivariogram analogue of the usual sample variance estimate s^2. While very natural, there is reason to doubt that this is the best estimate of the semivariogram. Certainly it will be sensitive to outliers, and the sample average of the squared differences may be rather badly behaved since under a Gaussian distributional assumption for the $Y(\mathbf{s}_i)$, the squared differences will have a distribution that is a scale multiple of the heavily skewed χ^2_1 distribution. In this regard, Cressie and Hawkins (1980) proposed a robustified estimate that uses sample averages of $|Y(\mathbf{s}_i) - Y(\mathbf{s}_j)|^{1/2}$; this estimate is available in several software packages (see Section 2.3 below). Perhaps more uncomfortable is the fact that (2.9) uses data differences, rather than the data itself. Also of concern is the fact that the components of the sum in (2.9) will be dependent within and across bins, and that $N(d_k)$ will vary across bins.

In any case, an empirical semivariogram estimate can be plotted, viewed, and an appropriately shaped theoretical variogram model can be fitted to this "data." Since any empirical estimate naturally carries with it a significant amount of noise in addition to its signal, this fitting of a theoretical model has traditionally been as much art as science: in any given real data setting, any number of different models (exponential, Gaussian, spherical, etc.) may seem equally appropriate. Indeed, fitting has historically been done "by eye," or at best by using trial and error to choose values of nugget, sill, and range parameters that provide a good match to the empirical semivariogram (where the "goodness" can be judged visually or by using some least squares or similar criterion); again see Section 2.3. More formally, we could treat this as a statistical estimation problem, and use nonlinear maximization routines to find nugget, sill, and range parameters that minimize some goodness-of-fit criterion.

If we also have a distributional model for the data, we could use maximum likelihood (or restricted maximum likelihood, REML) to obtain sensible parameter estimates; see, e.g., Smith (2001) for details in the case of Gaussian data modeled with the various parametric

variogram families outlined in Subsection 2.1.3. In Chapter 5 and Chapter 6 we shall see that the hierarchical Bayesian approach is broadly similar to this latter method, although it will often be easier and more intuitive to work directly with the covariance model $C(d)$, rather than changing to a partial likelihood in order to introduce the semivariogram. In addition, we will gain full inference, e.g., posterior distributions for all unknowns of interest as well as more accurate assessment of uncertainty than appealing to arguably inappropriate asymptotics (see Chapter 5).

2.2 Anisotropy

Stationary correlation functions extend the class of correlation functions from isotropy where association only depends upon distance to association that depends upon the separation vector between locations. As a result, association depends upon direction. Here, we explore covariance functions that are stationary but not isotropic. (We defer discussion of nonstationary covariance functions to Chapter 3.) A simple example is the class where we *separate* the components. That is, suppose we write the components of $\mathbf{s}$ as s_{lat}, s_{lon}, similarly for $\mathbf{h}$. Then, we can define $\text{corr}(Y(\mathbf{s}+\mathbf{h}), Y(\mathbf{s})) = \rho_1(h_{lat})\rho_2(h_{lon})$ where ρ_1 and ρ_2 are valid correlation functions on $\Re^1$. Evidently, this correlation function is stationary but depends on direction. In particular, if we switch h_{lat} and h_{lon} we will get a different value for the correlation even though $||\mathbf{h}||$ is unchanged. A common choice is $e^{\phi_1(|h_{lat}|)+\phi_2(|h_{lon}|)}$.

The separable correlation function is usually extended to a covariance function by introducing the multiplier σ^2, as we have done above. This covariance function tends to be used, for instance, with Gaussian processes in computer model settings (Santner, Williams and Notz, 2003; Rasmussen and Williams, 2006; Oakley and O'Hagan, 2004; Kennedy and O'Hagan, 2001). Here we seek a response surface over covariate space and, since the covariates live on their own spaces with their own scales, component-wise dependence seems appropriate. In the spatial setting, since lat and lon are on the same scale, it may be less suitable.

2.2.1 Geometric anisotropy

A commonly used class of stationary covariance functions is the geometric anisotropic covariance functions where we set

$$C(\mathbf{s}-\mathbf{s}') = \sigma^2\rho((\mathbf{s}-\mathbf{s}')^T B(\mathbf{s}-\mathbf{s}')) \; . \tag{2.10}$$

In (2.10), B is positive definite with ρ a valid correlation function in $\Re^r$ (say, from Table 2.1). We would omit the range/decay parameter since it can be incorporated into B. When $r = 2$ we obtain a specification with three parameters rather than one. Contours of constant association arising from c in (2.10) are elliptical. In particular, the contour corresponding to $\rho = .05$ provides the range in each spatial direction. Ecker and Gelfand (1997) provide the details for Bayesian modeling and inference incorporating (2.10); see also Subsection 6.1.4.

Following the discussion in Subsection 3.1.2, we can extend geometric anisotropy to *product* geometric anisotropy. In the simplest case, we would set

$$C(\mathbf{s}-\mathbf{s}') = \sigma^2 \; \rho_1((\mathbf{s}-\mathbf{s}')^T B_1(\mathbf{s}-\mathbf{s}')) \; \rho_2((\mathbf{s}-\mathbf{s}')^T B_2(\mathbf{s}-\mathbf{s}')) \; , \tag{2.11}$$

noting that c is valid since it arises as a product of valid covariance functions. See Ecker and Gelfand (2003) for further details and examples. Evidently, we can extend to a product of more than two geometric anisotropy forms and we can create rich directional range behavior. However, a challenge with (2.11) is that it introduces 7 parameters into the covariance function and it will be difficult to identify and learn about all of them unless we have many, many locations.

2.2.2 Other notions of anisotropy

In a more general discussion, Zimmerman (1993) suggests three different notions of anisotropy: *sill* anisotropy, *nugget* anisotropy, and *range* anisotropy. More precisely, working with a variogram $\gamma(\mathbf{h})$, let $\mathbf{h}$ be an arbitrary separation vector so that $\mathbf{h}/\|\mathbf{h}\|$ is a unit vector in $\mathbf{h}$'s direction. Consider $\gamma(a\mathbf{h}/\|\mathbf{h}\|)$. Let $a \to \infty$ and suppose $\lim_{a\to\infty} \gamma(a\mathbf{h}/\|\mathbf{h}\|)$ depends upon $\mathbf{h}$. This situation is naturally referred to as sill anisotropy. If we work with the usual relationship $\gamma(a\mathbf{h}/\|\mathbf{h}\|) = \tau^2 + \sigma^2\left(1 - \rho\left(a\frac{\mathbf{h}}{\|\mathbf{h}\|}\right)\right)$, then, in some directions, ρ must not go to 0 as $a \to \infty$. If this can be the case, then ergodicity assumptions (i.e., convergence assumptions associated with averaging) will be violated. If so, then perhaps the constant mean assumption, implicit for the variogram, does not hold. Alternatively, it is also possible that the constant nugget assumption fails.

Instead, let $a \to 0$ and suppose $\lim_{a\to 0} \gamma(a\mathbf{h}/\|\mathbf{h}\|)$ depends upon $\mathbf{h}$. This situation is referred to as nugget anisotropy. Since, by definition, ρ must go to 1 as $a \to 0$, this case says that the assumption of uncorrelated measurement errors with common variance may not be appropriate. In particular, a simple white noise process model with constant nugget for the nonspatial errors is not appropriate.

A third type of anisotropy is range anisotropy where the range depends upon direction. Zimmerman (1993) asserts that "this is the form most often seen in practice." Geometric anisotropy and the more general product geometric anisotropy from the previous subsections are illustrative cases. However, given the various constructive strategies offered in Subsection 3.1.2 to create more general stationary covariance functions, we can envision nongeometric range anisotropy, implying general correlation function or variogram contours in $\Re^2$. However, due to the positive definiteness restriction on the correlation function, the extent of possible contour shapes is still rather limited.

Lastly, motivated by directional variograms (see Subsection 2.3.2), some authors propose the idea of nested models (see Zimmerman, 1993, and the references therein). That is, for each separation vector there is an associated angle with, say, the x-axis, which by symmetry considerations can be restricted to $[0, \pi)$. Partitioning this interval into a set of angle classes, a different variogram model is assumed to operate for each class. In terms of correlations, this would imply a different covariance function is operating for each angle class. But evidently this does not define a valid process model: the resulting covariance matrix for an arbitrary set of locations need not be positive definite.

This can be seen with as few as three points and two angle classes. Let $(\mathbf{s}_1, \mathbf{s}_2)$ belong to one angle class with $(\mathbf{s}_1, \mathbf{s}_3)$ and $(\mathbf{s}_2, \mathbf{s}_3)$ in the other. With exponential isotropic correlation functions in each class by choosing ϕ_1 and ϕ_2 appropriately we can make $\rho(\mathbf{s}_1 - \mathbf{s}_2) \approx 0$ while $\rho(\mathbf{s}_1 - \mathbf{s}_3) = \rho(\mathbf{s}_2 - \mathbf{s}_3) \approx 0.8$. A quick calculation shows that the resulting 3×3 covariance (correlation) matrix is not positive definite. So, in terms of being able to write proper joint distributions for the resulting data, nested models are inappropriate; they do not provide an extension of isotropy that allows for likelihood based inference.

2.3 Exploratory approaches for point-referenced data

2.3.1 Basic techniques

Exploratory data analysis (EDA) tools are routinely implemented in the process of analyzing one- and two-sample data sets, regression studies, generalized linear models, etc. (see, e.g., Chambers et al., 1983; Hoaglin, Mosteller, and Tukey, 1983, 1985; Aiktin et al., 1989). Similarly, such tools are appropriate for analyzing point-referenced spatial data.

For continuous spatial data, the starting point is the so-called "first law of geostatistics." Figure 2.2 illustrates this "law" in a one-dimensional setting. The data is partitioned into a mean term and an error term. The mean corresponds to global (or *first-order*) behavior,

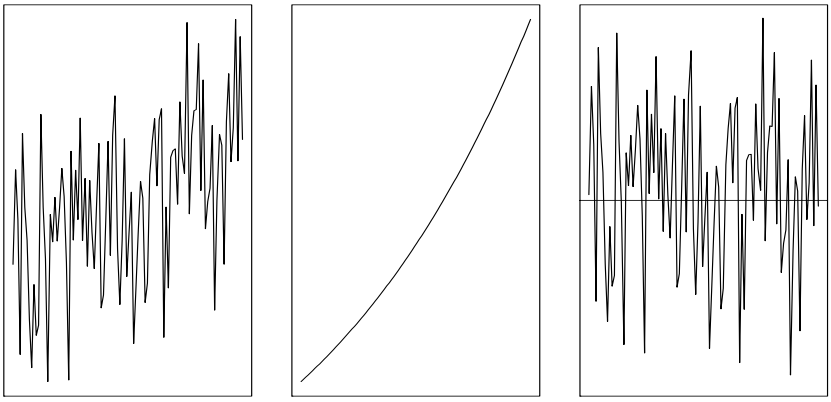


Figure 2.2 *Illustration of the first law of geostatistics.*

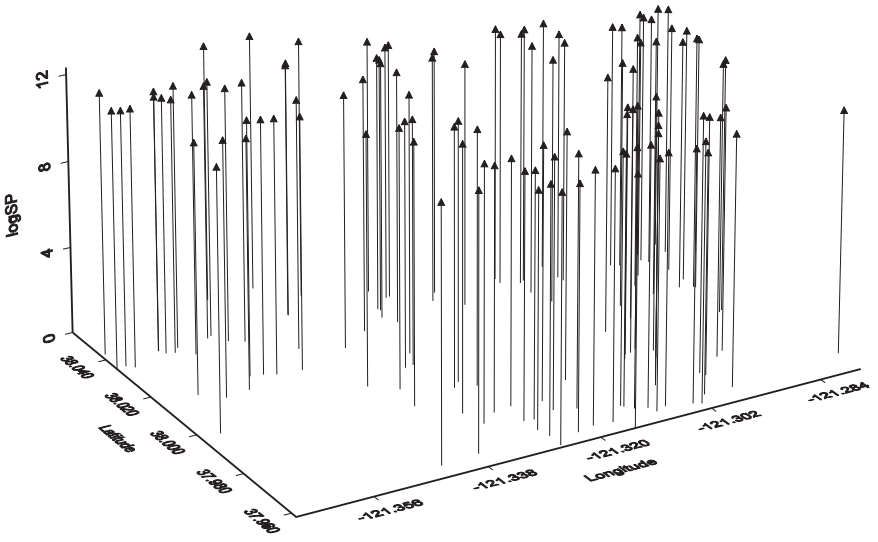


Figure 2.3 *Illustrative three-dimensional "drop line" scatterplot, Stockton data.*

while the error captures local (or *second-order*) behavior through a covariance function. EDA tools are available to examine both first- and second-order behavior.

The law also clarifies that spatial association in the data, $Y(\mathbf{s})$, need not resemble spatial association in the residuals, $\epsilon(\mathbf{s})$. That is, spatial association in the $Y(\mathbf{s})$ corresponds to looking at $E(Y(\mathbf{s}) - \mu)(Y(\mathbf{s}') - \mu)$, while spatial structure in the $\epsilon(\mathbf{s})$ corresponds to looking at $E(Y(\mathbf{s}) - \mu(\mathbf{s}))(Y(\mathbf{s}') - \mu(\mathbf{s}'))$. The difference between the former and the latter is $(\mu - \mu(\mathbf{s}))(\mu - \mu(\mathbf{s}'))$, which, if interest is in spatial regression, we would not expect to be negligible.

Certainly an initial exploratory display should be a simple map of the locations themselves. We need to assess how *regular* the arrangement of the points is and also whether there is a much larger maximum distance between points in some directions than in others. Next, some authors would recommend a stem-and-leaf display of the $Y(\mathbf{s})$. This plot is evidently nonspatial and is customarily for observations which are i.i.d. We expect both nonconstant mean and spatial dependence, but such a plot may at least suggest potential outliers. Next

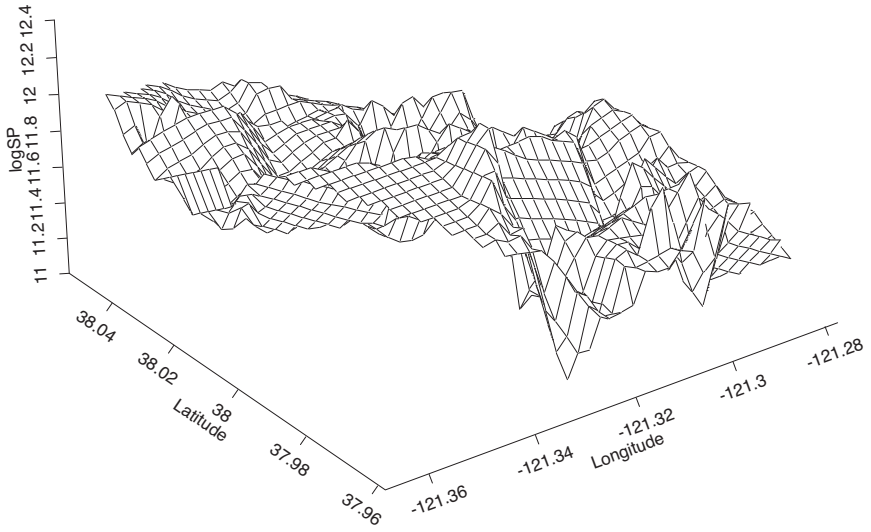


Figure 2.4 *Illustrative three-dimensional surface ("perspective") plot, Stockton real estate data.*

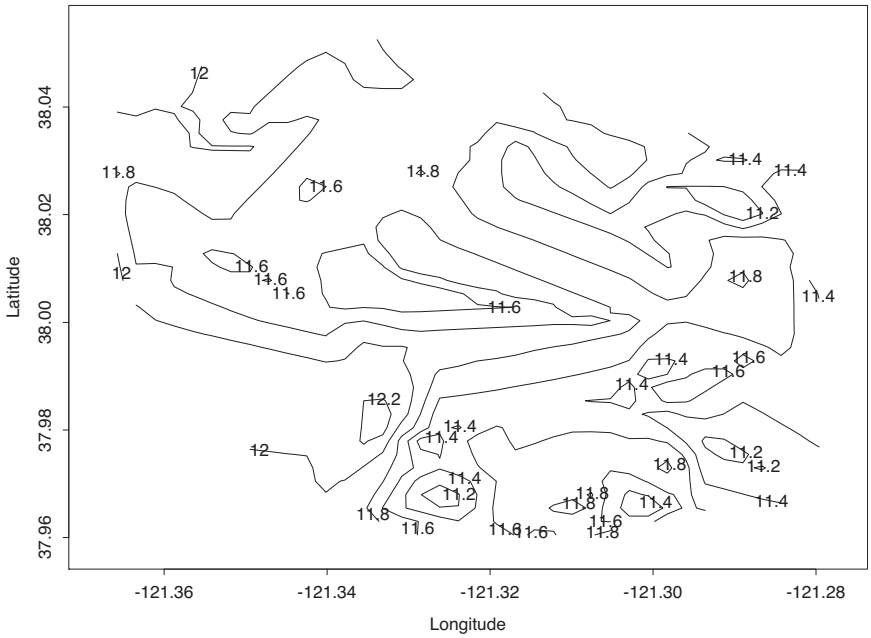


Figure 2.5 *Illustrative contour plot, Stockton real estate data.*

we might develop a three-dimensional "drop line" scatterplot of $Y(\mathbf{s}_i)$ versus $\mathbf{s}_i$, alternatively, a three-dimensional surface plot or perhaps, a contour plot as a *smoothed* summary. Examples of these three plots are shown for a sample of 120 log-transformed home selling prices in Stockton, CA, in Figures 2.3, 2.4, and 2.5, respectively. Of these three, we find the contour plot to be the most effective in revealing the entire spatial surface. However, as the preceding paragraph clarifies, such displays may be deceiving. They may show spatial pattern that will disappear after $\mu(\mathbf{s})$ is fitted, or perhaps vice versa. It seems more sensible to study spatial pattern in the residuals.

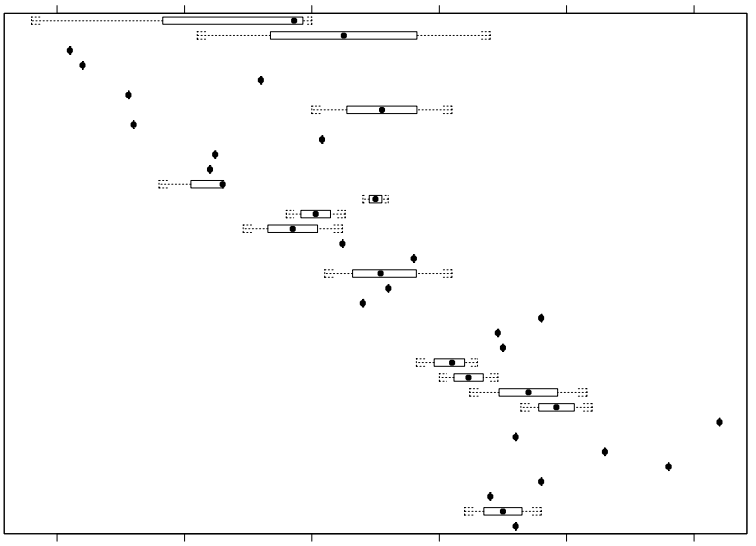


Figure 2.6 *Illustrative row box plots, Diggle and Ribeiro (2002) surface elevation data.*

In exploring $\mu(\mathbf{s})$ we may have two types of information at location $\mathbf{s}$. One is the purely geographic information, i.e., the geocoded location expressed in latitude and longitude or as projected coordinates such as eastings and northings (Subsection 1.2.1 above). The other will be features relevant for explaining the $Y(\mathbf{s})$ at $\mathbf{s}$. For instance, if $Y(\mathbf{s})$ is a pollution concentration, then elevation, temperature, and wind information at $\mathbf{s}$ could well be useful and important. If, instead, $Y(\mathbf{s})$ is the selling price of a single-family home at $\mathbf{s}$, then characteristics of the home (square feet, age, number of bathrooms, etc.) would be useful.

When the mean is described purely through geographic information, $\mu(\mathbf{s})$ is referred to as a *trend surface*. When $\mathbf{s} \in \Re^2$, the surface is usually developed as a low-dimensional bivariate polynomial. For data that is roughly gridded (or can be assigned to row and column bins by overlaying a regular lattice on the points), we can make row and column boxplots looking for trend. Displaying these boxplots versus their center could clarify the existence and nature of such trend. In fact, median polishing (see, e.g., Hoaglin, Mosteller, and Tukey, 1985) could be used to extract row and column effects, and also to see if a multiplicative trend surface term is useful; see Cressie (1983, pp. 46–48) in this regard.

Figures 2.6 and 2.7 illustrate the row and column boxplot approach for a data set previously considered by Diggle and Ribeiro (2002). The response variable is the surface elevation ("height") at 52 locations on a regular grid within a 310-foot square (and where the mesh of the grid is 50 feet). The plots reveals some evidence of spatial pattern as we move along the rows, but not down the columns of the regular grid.

To assess small-scale behavior, some authors recommend creating the *semivariogram cloud*, i.e., a plot of $(Y(\mathbf{s}_i) - Y(\mathbf{s}_j))^2$ versus $||\mathbf{s}_i - \mathbf{s}_j||$. Usually this cloud is too "noisy" to reveal very much; see, e.g., Figure 6.1. The empirical semivariogram (2.9) is preferable in terms of reducing some of the noise, and can be a helpful tool in assessing the presence of spatial structure. Again, the caveat above suggests employing it for residuals (not the data itself) unless a constant mean is appropriate.

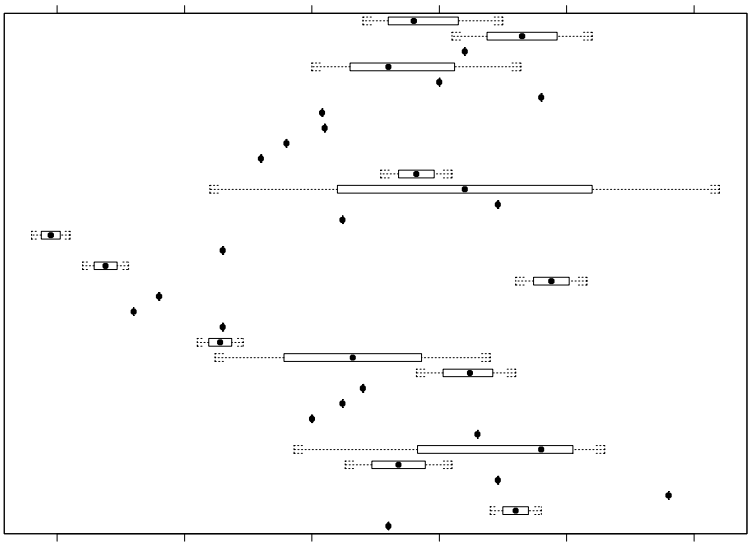


Figure 2.7 *Illustrative column box plots, Diggle and Ribeiro (2002) surface elevation data.*

An empirical (nonparametric) covariance estimate, analogous to (2.9), is also available. Creating bins as in this earlier approach, define

$$\widehat{C}(t_k) = \frac{1}{N_k} \sum_{(\mathbf{s}_i,\mathbf{s}_j)\in N(t_k)} (Y(\mathbf{s}_i) - \bar{Y})(Y(\mathbf{s}_j) - \bar{Y}) \ , \tag{2.12}$$

where again $N(t_k) = \{(\mathbf{s}_i, \mathbf{s}_j) : ||\mathbf{s}_i - \mathbf{s}_j|| \in I_k\}$ for $k = 1, \ldots, K$, I_k indexes the kth bin, and there are N_k pairs of points falling in this bin. Equation (2.12) is a spatial generalization of a lagged autocorrelation in time series analysis. Since $\widehat{c}$ uses a common $\bar{Y}$ for all $Y(\mathbf{s}_i)$, it may be safer to employ (2.12) on the residuals. We note that the empirical covariance function is not used much in practice; the empirical variogram is much more common. Moreover, two further issues arise: first, how should we define $\widehat{C}(0)$, and second, regardless of this choice, we note that $\widehat{\gamma}(t_k)$ does *not* equal $\widehat{C}(0) - \widehat{C}(t_k)$, $k = 1, \ldots, K$. Details for both of these issues are left to Exercise 5. Again, since we are only viewing $\widehat{(\gamma)}$ and $\widehat{C}$ as exploratory tools, there seems to be no reason to pursue the latter further.

Again, with a regular grid or binning we can create "same-lag" scatterplots. These are plots of $Y(\mathbf{s}_i + h\mathbf{e})$ versus $Y(\mathbf{s}_i)$ for a fixed h and a fixed unit vector $\mathbf{e}$. Comparisons among such plots may reveal the presence of anisotropy and perhaps nonstationarity.

Lastly, suppose we attach a neighborhood to each point. We can then compute the sample mean and variance for the points in the neighborhood, and even a sample correlation coefficient using all pairs of data in the neighborhood. Plots of each of them versus location can be informative. The first may give some idea regarding how the mean structure changes across the study region. Plots of the second and third may provide evidence of nonstationarity. Implicit in extracting useful information from these plots is a roughly constant local mean. If $\mu(\mathbf{s})$ is to be a trend surface, this is plausible. But if $\mu(\mathbf{s})$ is a function of some geographic variables at $\mathbf{s}$ (say, home characteristics), then use of residuals would be preferable.

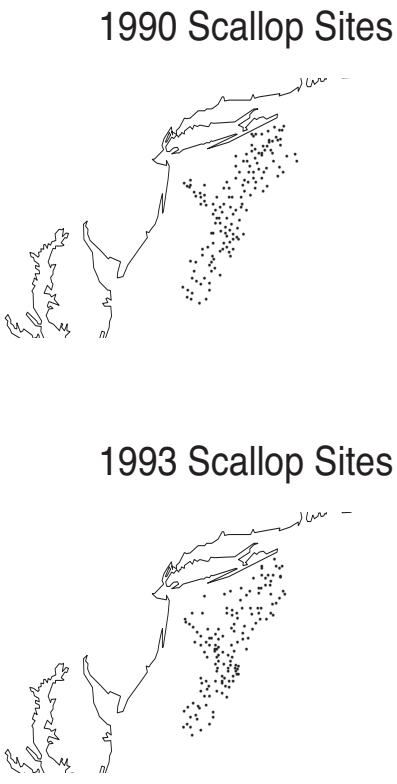


Figure 2.8 *Sites sampled in the Atlantic Ocean for 1990 and 1993 scallop catch data.*

2.3.2 Assessing anisotropy

We illustrate various EDA techniques to assess anisotropy using sampling of scallop abundance on the continental shelf off the coastline of the northeastern U.S. The data comes from a survey conducted by the Northeast Fisheries Science Center of the National Marine Fisheries Service. Figure 2.8 shows the sampling sites for 1990 and 1993. We see much more sampling in the southwest to northeast direction than in the northwest to southeast direction. Evidently, it is more appropriate to follow the coastline in searching for scallops.

2.3.2.1 Directional semivariograms and rose diagrams

The most common EDA technique for assessing anisotropy involves use of directional semivariograms. Typically, one chooses angle classes $\eta_i \pm \epsilon$, $i = 1, \ldots, L$ where ϵ is the halfwidth of the angle class and L is the number of angle classes. For example, a common choice of angle classes involves the four cardinal directions measured counterclockwise from the x-axis (0°, 45°, 90°, and 135°) where ϵ is 22.5°. Journel and Froidevaux (1982) display directional semivariograms at angles 35°, 60°, 125°, and 150° in deducing anistropy for a tungsten deposit. While knowledge of the underlying spatial characteristics of region D is invaluable in choosing directions, often the choice of the number of angle classes and the directions seems to be arbitrary.

For a given angle class, the Matheron empirical semivariogram (2.9) can be used to provide a directional semivariogram for angle η_i. Theoretically, all types of anisotropy can be assessed from these directional semivariograms; however, in practice determining whether the sill, nugget, and/or range varies with direction can be difficult. In particular, it is unclear how much variability will arise in directional variograms generated under

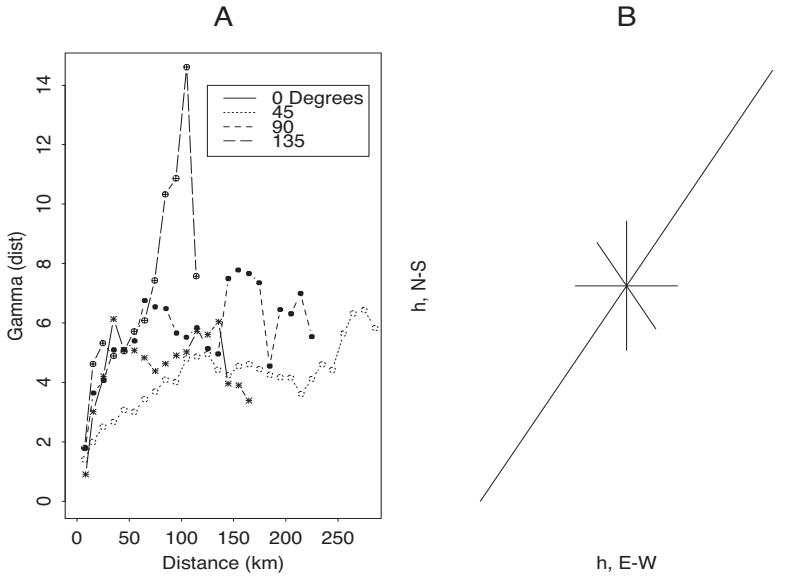


Figure 2.9 *Directional semivariograms (a) and a rose diagram (b) for the 1990 scallop data.*

isotropy. Figure 2.9(a) illustrates directional semivariograms for the 1990 scallop data in the four cardinal directions. Note that the semivariogram points are connected only to aid comparison.

Possible conclusions from the figure are: the variability in the 45° direction (parallel to the coastline) is significantly less than in the other three directions and the variability perpendicular to the coastline (135°) is very erratic, possibly exhibiting sill anisotropy. We caution however that it is dangerous to read too much significance and interpretation into directional variograms. For example, from Figure 2.8, as noted above, we see far more sampling at greater distances in the southwest-northeast direction than in the northwest-southeast direction. Moreover, no sample sizes (and thus no assessments of variability) are attached to these pictures. Directional variograms from data generated under a simple isotropic model will routinely exhibit differences of the magnitudes seen in Figure 2.9(a). Furthermore, it seems difficult to draw any conclusions regarding the presence of geometric anisotropy from this figure.

A rose diagram (Isaaks and Srivastava, 1989, pp. 151–154) can be created from the directional semivariograms to evaluate geometric anisotropy. At an arbitrarily selected γ^*, for a directional semivariogram at angle η, the distance d^* at which the directional semivariogram attains γ^* can be interpolated. Then, the rose diagram is a plot of angle η and corresponding distance d^* in polar coordinates. If an elliptical contour describes the extremities of the rose diagram reasonably well, then the process exhibits geometric anisotropy. For instance, the rose diagram for the 1990 scallop data is presented in Figure 2.9(b) using the γ^* contour of 4.5. It is approximately elliptical, oriented parallel to the coastline ($\approx 45°$) with a ratio of major to minor ellipse axes of about 4.

2.3.2.2 Empirical semivariogram contour (ESC) plots

A more informative method for assessing anisotropy is a contour plot of the empirical semivariogram surface in $\Re^2$. Such plots are mentioned informally in Isaaks and Srivastava (1989, pp. 149–151) and in Haining (1990, pp. 284–286); the former call them contour maps of the grouped variogram values, the latter an isarithmic plot of the semivariogram.

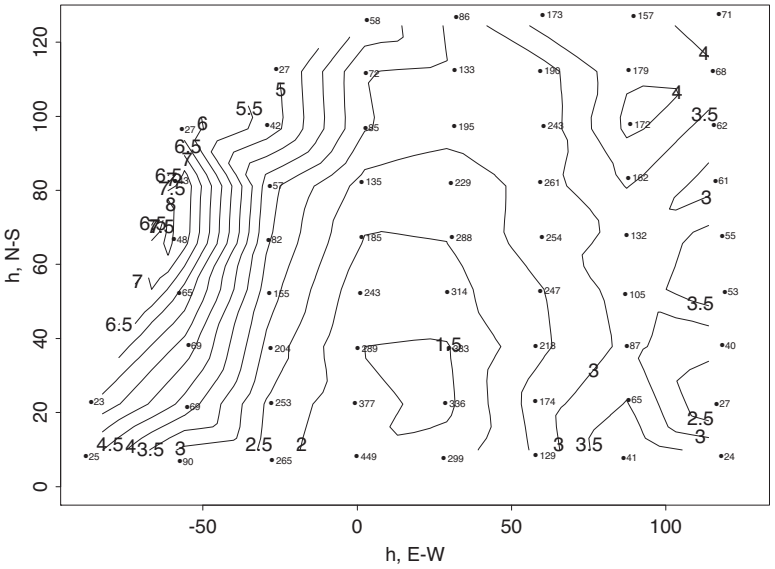


Figure 2.10 *ESC plot for the 1993 scallop data.*

Following Ecker and Gelfand (1999), we formalize such a plot here, calling it an *empirical semivariogram contour* (ESC) plot. For each of the $\frac{N(N-1)}{2}$ pairs of sites in $\Re^2$, calculate h_x and h_y, the separation distances along each axis. Since the sign of h_y depends upon the arbitrary order in which the two sites are compared, we demand that $h_y \geq 0$. (We could alternatively demand that $h_x \geq 0$.) That is, we take $(-h_x, -h_y)$ when $h_y < 0$. These separation distances are then aggregated into rectangular bins B_{ij} where the empirical semivariogram values for the (i, j)th bin are calculated by

$$\gamma^*_{ij} = \frac{1}{2N_{B_{ij}}} \sum_{\{(k,l):(\mathbf{s}_k - \mathbf{s}_l) \in B_{ij}\}} (Y(\mathbf{s}_k) - Y(\mathbf{s}_l))^2 \,, \tag{2.13}$$

where $N_{B_{ij}}$ equals the number of sites in bin B_{ij}. Because we force $h_y \geq 0$ with h_x unrestricted, we make the bin width on the y-axis half of that for the x-axis. We also force the middle class on the x-axis to be centered around zero. Upon labeling the center of the (i, j)th bin by (x_i, y_j), a three dimensional plot of γ^*_{ij} versus (x_i, y_j) yields an empirical semivariogram surface. Smoothing this surface using interpolation algorithms (e.g., Akima, 1978) produces a contour plot that we call the ESC plot. A symmetrized version of the ESC plot can be created by reflecting the upper left quadrant to the lower right and the upper right quadrant to the lower left.

The ESC plot can be used to assess departures from isotropy; isotropy is depicted by circular contours while elliptical contours capture geometric anisotropy. A rose diagram traces only one arbitrarily selected contour of this plot. A possible drawback to the ESC plot is the occurrence of sparse counts in extreme bins. However, these bins may be trimmed before smoothing if desired. Concerned that use of geographic coordinates could introduce artificial anisotropy (since 1° latitude $\neq$ 1° longitude in the northeastern United States), we have employed a Universal Transverse Mercator (UTM) projection to kilometers in the E-W and N-S axes (see Subsection 1.2.1).

Figure 2.10 is the empirical semivariogram contour plot constructed using x-axis width of 30 kilometers for the 1993 scallop data. We have overlaid this contour plot on the bin centers with their respective counts. Note that using empirical semivariogram values in the row of the ESC plot for which $h_y \approx 0$ provides an alternative to the usual 0° directional

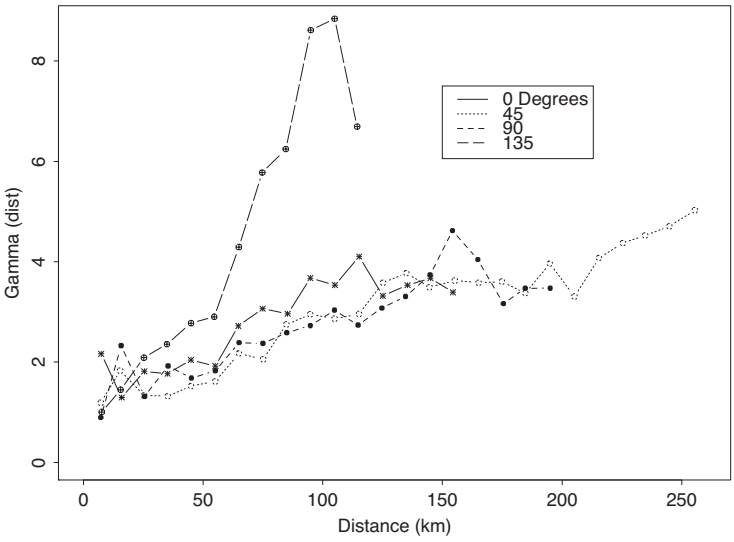


Figure 2.11 *Directional semivariograms for the 1993 scallop data.*

semivariogram. The latter directional semivariograms are based on a polar representation of the angle and distance. For a chosen direction η and tolerance ϵ, the area for a class fans out as distance increases (see Figure 7.1 of Isaaks and Srivastava, 1989, p. 142). Attractively, a directional semivariogram based on the rectangular bins associated with the empirical semivariogram in $\Re^2$ has bin area remaining constant as distance increases. In Figure 2.11, we present the four customary directional (polar representation) semivariograms for the 1993 scallop data. Clearly, the ESC plot is more informative, particularly in suggesting evidence of geometric anisotropy.

2.4 Classical spatial prediction

In this section we describe the classical (i.e., minimum mean-squared error) approach to spatial prediction in the point-referenced data setting. The approach is commonly referred to as *kriging*, so named by Matheron (1963) in honor of D.G. Krige, a South African mining engineer whose seminal work on empirical methods for geostatistical data (Krige, 1951) inspired the general approach (and indeed, inspired the convention of using the terms "point-level spatial analysis" and "geostatistical analysis" interchangeably!). The problem is one of optimal spatial prediction: given observations of a random field $\mathbf{Y} = (Y(\mathbf{s}_1), \ldots, Y(\mathbf{s}_n))'$, how do we predict the variable Y at a site $\mathbf{s}_0$ where it has not been observed? In other words, what is the best predictor of the value of $Y(\mathbf{s}_0)$ based upon the data $\mathbf{y}$?

A linear predictor for $Y(\mathbf{s}_0)$ based on $\mathbf{y}$ would take the form $\sum \ell_i Y(\mathbf{s}_i) + \delta_0$. Using squared error loss, the best linear prediction would minimize $E[Y(\mathbf{s}_0) - (\sum \ell_i Y(\mathbf{s}_i) + \delta_0)]^2$ over δ_0 and the ℓ_i. Under the intrinsic stationarity specification (2.1) we would take $\sum \ell_i = 1$ in order that $E(Y(\mathbf{s}_0) - \sum \ell_i Y(\mathbf{s}_i)) = 0$. As a result, we would minimize $E[Y(\mathbf{s}_0) - \sum \ell_i Y(\mathbf{s}_i)]^2 + \delta_0^2$, and clearly δ_0 would be set to 0. Now letting $a_0 = 1$ and $a_i = -\ell_i$ we see that the criterion becomes $E[\sum_{i=0}^n a_i Y(\mathbf{s}_i)]^2$ with $\sum a_i = 0$. But from (2.4) this expectation becomes $-\sum_i \sum_j a_i a_j \gamma(\mathbf{s}_i - \mathbf{s}_j)$, revealing how, historically, the variogram arose in kriging within the geostatistical framework. Indeed, the optimal ℓ's can be obtained by solving this constrained optimization. In fact, it is a routine optimization of a quadratic from under a linear constraint and is customarily handled using Lagrange multipliers. The solution will be a function of $\gamma(\mathbf{h})$, in fact of the set of $\gamma_{ij} = \gamma(\mathbf{s}_i - \mathbf{s}_j)$ and $\gamma_{0j} = \gamma(\mathbf{s}_0 - \mathbf{s}_j)$ With an

estimate of γ, one immediately obtains the so-called *ordinary kriging* estimate. Other than the intrinsic stationarity model (Subsection 2.1.2), no further distributional assumptions are required for the $Y(\mathbf{s})$'s.

To provide a bit more detail, restoring the ℓ_i's, we obtain

$$-\sum_i \sum_j a_i a_j \gamma(\mathbf{s}_i - \mathbf{s}_j) = -\sum_i \sum_j \ell_i \ell_j \gamma_{ij} + 2\sum_i \ell_i \gamma_{0i}$$

Adding the constraint, $\sum \ell_i = 1$, times the Lagrange multiplier, λ, we find that the partial derivative of this expression with regard to ℓ_i becomes $-\sum_j \ell_j \gamma_{ij} + \gamma_{0i} - \lambda = 0$. Letting Γ be the $n \times n$ matrix with entries $\Gamma_{ij} = \gamma_{ij}$ and $\boldsymbol{\gamma}_0$ be the $n \times 1$ vector with $(\boldsymbol{\gamma}_0)_i = \gamma_{0i}$, with $\boldsymbol{\ell}$ the $n \times 1$ vector of coefficients, we obtain the system of *kriging* equations, $\Gamma\boldsymbol{\ell} + \lambda\mathbf{1} = \boldsymbol{\gamma}_0$ and $\mathbf{1}^T\boldsymbol{\ell} = 1$. The solution is $\boldsymbol{\ell} = \Gamma^{-1}\left(\boldsymbol{\gamma}_0 + \frac{(1 - \mathbf{1}^T\Gamma^{-1}\boldsymbol{\gamma}_0)}{\mathbf{1}^T\Gamma^{-1}\mathbf{1}}\mathbf{1}\right)$ and $\boldsymbol{\ell}^T\mathbf{Y}$ becomes the best linear unbiased predictor (BLUP). Again, with $\gamma(\mathbf{h})$ unknown, we have to estimate it in order to calculate this estimator. Then, $\boldsymbol{\ell}$ is a function of the data and the estimator is no longer linear. Continuing in this regard, the usual estimator of the uncertainty in the prediction is the predictive mean square error (PMSE), $E(Y(\mathbf{s}_0) - \boldsymbol{\ell}^T\mathbf{Y})^2$, rather than $\text{var}(\boldsymbol{\ell}^T\mathbf{Y})$. There is a closed form expression for the former, i.e., $\boldsymbol{\gamma}_0{}^T\Gamma^{-1}\boldsymbol{\gamma}_0 - (\boldsymbol{\ell}^T\Gamma^{-1}\boldsymbol{\gamma}_0 - 1)^2/\boldsymbol{\ell}^T\Gamma^{-1}\boldsymbol{\ell}$, which, also requires an estimate of $\gamma(\mathbf{h})$ in order to be calculated.

Let us now take a formal look at kriging in the context of Gaussian processes. Consider first the case where we have no covariates, but only the responses $Y(\mathbf{s}_i)$. This is developed by means of the following model for the observed data:

$$\mathbf{Y} = \mu\mathbf{1} + \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} \sim N(\mathbf{0}, \Sigma) \ .$$

For a spatial covariance structure having no nugget effect, we specify Σ as

$$\Sigma = \sigma^2 H(\phi) \text{ where } (H(\phi))_{ij} = \rho(\phi; d_{ij}) \ ,$$

where $d_{ij} = ||\mathbf{s}_i - \mathbf{s}_j||$, the distance between $\mathbf{s}_i$ and $\mathbf{s}_j$ and ρ is a valid correlation function on $\Re^r$ such as those in Table 2.1. For a model having a nugget effect, we instead set

$$\Sigma = \sigma^2 H(\phi) + \tau^2 I \ ,$$

where τ^2 is the nugget effect variance.

When covariate values $\mathbf{x} = (x(\mathbf{s}_1), \ldots, x(\mathbf{s}_n))'$ and $x(\mathbf{s}_0)$ are available for incorporation into the analysis, the procedure is often referred to as *universal kriging*, though we caution that some authors (e.g., Kaluzny et al., 1998) use the term "universal" in reference to the case where only latitude and longitude are available as covariates. The model now takes the more general form

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} \sim N(\mathbf{0}, \Sigma) \ ,$$

with Σ being specified as above, either with or without the nugget effect. Note that ordinary kriging may be looked upon as a particular case of universal kriging with X being the $n \times 1$ matrix (i.e., column vector) $\mathbf{1}$, and $\boldsymbol{\beta}$ the scalar μ.

We now pose our prediction problem as follows: we seek the function $h(\mathbf{y})$ that minimizes the mean-squared prediction error,

$$E\left[(Y(\mathbf{s}_0) - h(\mathbf{y}))^2 \,\middle|\, \mathbf{y}\right] \ . \tag{2.14}$$

By adding and subtracting the conditional mean $E[Y(\mathbf{s}_0)|\mathbf{y}]$ inside the square, grouping terms, and squaring we obtain

$$\begin{aligned} &E\left[(Y(\mathbf{s}_0) - h(\mathbf{y}))^2 \,\middle|\, \mathbf{y}\right] \\ &\quad = E\left\{(Y(\mathbf{s}_0) - E[Y(\mathbf{s}_0)|\mathbf{y}])^2 \,\middle|\, \mathbf{y}\right\} + \{E[Y(\mathbf{s}_0)|\mathbf{y}] - h(\mathbf{y})\}^2 \ , \end{aligned}$$

since (as often happens in statistical derivations like this) the expectation of the cross-product term equals zero. But, since the second term on the right-hand side is nonnegative, we have

$$E\left[(Y(\mathbf{s}_0) - h(\mathbf{y}))^2 \,\middle|\, \mathbf{y}\right] \geq E\left\{(Y(\mathbf{s}_0) - E[Y(\mathbf{s}_0)|\mathbf{y}])^2 \,\middle|\, \mathbf{y}\right\}$$

for any function $h(\mathbf{y})$. Equality holds if and only if $h(\mathbf{y}) = E[Y(\mathbf{s}_0)|\mathbf{y}]$, so it must be that the predictor $h(\mathbf{y})$ which minimizes the error is the conditional expectation of $Y(\mathbf{s}_0)$ given the data. This result is familiar since we know that the mean minimizes expected squared error loss. From a Bayesian point of view, this $h(\mathbf{y})$ is just the *posterior mean* of $Y(\mathbf{s}_0)$, and it is well known that the posterior mean is the Bayes rule (i.e., the minimizer of posterior risk) under squared error loss functions of the sort adopted in (2.14) above as our scoring rule. Note that the posterior mean is the best predictor under squared error loss regardless of whether we assume a Gaussian model and that, in general, it need not be linear; it need not be the BLUP which is the ordinary kriging predictor. However, under a Gaussian process model assumption, this posterior mean is linear, as we now clarify.

In particular, let us explicitly obtain the form of the posterior mean of $Y(\mathbf{s}_0)$. It will emerge as a function of all the population parameters ($\boldsymbol{\beta}, \sigma^2, \phi$, and τ^2). From standard multivariate normal theory we have the following general result: If

$$\begin{pmatrix} \mathbf{Y}_1 \\ \mathbf{Y}_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{pmatrix}\right),$$

where $\Omega_{21} = \Omega_{12}^T$, then the conditional distribution $p(\mathbf{Y}_1|\mathbf{Y}_2)$ is normal with mean and variance:

$$\begin{aligned} E[\mathbf{Y}_1|\mathbf{Y}_2] &= \boldsymbol{\mu}_1 + \Omega_{12}\Omega_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2); \\ Var[\mathbf{Y}_1|\mathbf{Y}_2] &= \Omega_{11} - \Omega_{12}\Omega_{22}^{-1}\Omega_{21}\,. \end{aligned}$$

In our framework, we have $\mathbf{Y}_1 = Y(\mathbf{s}_0)$ and $\mathbf{Y}_2 = \mathbf{y}$. It then follows that

$$\Omega_{11} = \sigma^2 + \tau^2, \quad \Omega_{12} = \boldsymbol{\gamma}^T, \quad \text{and } \Omega_{22} = \Sigma = \sigma^2 H(\phi) + \tau^2 I\,,$$

where $\boldsymbol{\gamma}^T = \left(\sigma^2\rho(\phi; d_{01}), \ldots, \sigma^2\rho(\phi; d_{0n})\right)$. Substituting these values into the mean and variance formulae above, we obtain

$$E[Y(\mathbf{s}_0)|\mathbf{y}] = \mathbf{x}_0^T\boldsymbol{\beta} + \boldsymbol{\gamma}^T\Sigma^{-1}(\mathbf{y} - X\boldsymbol{\beta})\,, \tag{2.15}$$

$$\text{and } Var[Y(\mathbf{s}_0)|\mathbf{y}] = \sigma^2 + \tau^2 - \boldsymbol{\gamma}^T\Sigma^{-1}\boldsymbol{\gamma}\,. \tag{2.16}$$

We see that the posterior mean is a linear predictor. We remark that this solution assumes we have actually observed the covariate value $\mathbf{x}_0 = \mathbf{x}(\mathbf{s}_0)$ at the "new" site $\mathbf{s}_0$; we defer the issue of missing $\mathbf{x}_0$ for the moment.

Since, in practice, the model parameters are unknown, they must be estimated from the data. Here we would modify $h(\mathbf{y})$ to

$$\widehat{h(\mathbf{y})} = \mathbf{x}_0^T\widehat{\boldsymbol{\beta}} + \widehat{\boldsymbol{\gamma}}^T\widehat{\Sigma}^{-1}\left(\mathbf{y} - X\widehat{\boldsymbol{\beta}}\right)\,,$$

where $\widehat{\boldsymbol{\gamma}} = \left(\hat{\sigma}^2\rho(\hat{\phi}; d_{01}), \ldots, \hat{\sigma}^2\rho(\hat{\phi}; d_{0n})\right)^T$, $\widehat{\boldsymbol{\beta}} = \left(X^T\widehat{\Sigma}^{-1}X\right)^{-1}X^T\widehat{\Sigma}^{-1}\mathbf{y}$, the usual weighted least squares estimator of $\boldsymbol{\beta}$, and $\widehat{\Sigma} = \hat{\sigma}^2 H(\hat{\phi})$. Thus $\widehat{h(\mathbf{y})}$ can be written as $\boldsymbol{\lambda}^T\mathbf{y}$, where

$$\boldsymbol{\lambda} = \widehat{\Sigma}^{-1}\widehat{\boldsymbol{\gamma}} + \widehat{\Sigma}^{-1}X\left(X^T\widehat{\Sigma}^{-1}X\right)^{-1}\left(\mathbf{x}_0 - X^T\widehat{\Sigma}^{-1}\widehat{\boldsymbol{\gamma}}\right)\,. \tag{2.17}$$

The reader may be curious as to whether $\boldsymbol{\lambda}^T\mathbf{Y}$ is the same as $\boldsymbol{\ell}^T\mathbf{Y}$ assuming all parameters are known say, in the ordinary kriging case. That is, assuming we have $\gamma(\mathbf{h}) = c(\mathbf{0}) - c(\mathbf{h})$, we can write both linear predictors in terms of $c(\mathbf{h})$ or $\gamma(\mathbf{h})$ but will they agree? Immediately, we know that the answer is no because $\boldsymbol{\ell}^T\mathbf{Y}$ depends only on $\gamma(\mathbf{h})$ or $c(\mathbf{h})$ while $\boldsymbol{\lambda}^T\mathbf{Y}$ requires $\boldsymbol{\beta}$, hence μ in the constant mean case. In fact, it is a straightforward calculation to show that, if we replace μ by $\hat{\mu}$, the best linear unbiased estimator of μ, in $\boldsymbol{\lambda}$, we do obtain the ordinary kriging estimator. We leave this as an exercise.

If $\mathbf{x}_0$ is unobserved, we can estimate it and $Y(\mathbf{s}_0)$ jointly by iterating between this formula and a corresponding one for $\hat{\mathbf{x}}_0$, namely

$$\hat{\mathbf{x}}_0 = X^T\boldsymbol{\lambda}\,,$$

which arises simply by multiplying both sides of (2.17) by X^T and simplifying. This is essentially an EM (expectation-maximization) algorithm (Dempster, Laird, and Rubin, 1977), with the calculation of $\hat{\mathbf{x}}_0$ being the E step and (2.17) being the M step.

In the classical framework a lot of energy is devoted to the determination of the optimal estimates to plug into the above equations. Typically, restricted maximum likelihood (REML) estimates are selected and shown to have certain optimal properties. However, as we shall see in Chapter 6, how to perform the estimation is not an issue in the Bayesian setting. There, we instead impose prior distributions on the parameters and produce the full posterior predictive distribution $p\,(Y(\mathbf{s}_0)|\mathbf{y})$. Any desired point or interval estimate (the latter to express our uncertainty in such prediction), as well as any desired probability statements, may then be computed with respect to this distribution.

2.4.0.3 Noiseless kriging

It is natural to ask whether the kriging predictor (ordinary or posterior mean) will return the observed value at the $\mathbf{s}_i$'s where we observed the process. If so, we would refer to the predictor as an *exact* interpolator. The literature addresses this problem through detailed inspection of the kriging equations. However, the answer is immediately clear once we look at the model specification. And, it is possibly counterintuitive; we obtain exact interpolation in the case of no nugget, rather than in the seemingly more flexible case where we add a nugget.

Analytically, one can determine whether or not the predictor in (2.15) will equal the observed value at a given $\mathbf{s}_i$. We leave as a formal exercise to verify that if $\tau^2 = 0$ (i.e., the no-nugget case, or so-called noiseless prediction), then the answer is yes, while if $\tau^2 > 0$ then the answer is no.

However, we can illuminate the situation without formal calculation. There are two potential settings: (i) $Y(\mathbf{s}) = \mu(\mathbf{s}) + w(\mathbf{s})$ and (ii) $Y(\mathbf{s}) = \mu(\mathbf{s}) + w(\mathbf{s}) + \epsilon(\mathbf{s})$, the no-nugget and nugget cases, respectively. Under (i), evidently, predicting $Y(\mathbf{s}_0)$ is the same as predicting $\mu(\mathbf{s}_0) + w(\mathbf{s}_0)$. There is only one surface for the process realization. Under (ii), predicting $Y(\mathbf{s}_0)$ is different from predicting $\mu(\mathbf{s}_0) + w(\mathbf{s}_0)$; we have two random surfaces where the former is everywhere discontinuous while the latter, for customary covariance functions, is continuous. Here, terminology can be confusing. Under (i), there is no noise, so we could refer to this case as noiseless kriging. However, under (ii), predicting $\mu(\mathbf{s}_0) + w(\mathbf{s}_0)$ could be referred to as noiseless interpolation. We ignore the terminology issue and address the exact interpolation question.

Now, suppose $\mathbf{s}_0$ is, in fact, one of the $\mathbf{s}_i$. Then, we need to distinguish the observed value at $\mathbf{s}_i$ say $Y_{obs}(\mathbf{s}_i)$ from a new or replicate value at $\mathbf{s}_i$, say $Y_{rep}(\mathbf{s}_i)$. Under (i) $f(Y_{rep}(\mathbf{s}_i)|\text{Data}) = f(\mu(\mathbf{s}_i) + w(\mathbf{s}_i)|\text{Data})$ and since, given the data, $\mu(\mathbf{s}_i) + w(\mathbf{s}_i) = Y_{obs}(\mathbf{s}_i)$, $f(Y_{rep}(\mathbf{s}_i)|\text{Data})$ is degenerate at $Y_{obs}(\mathbf{s}_i)$; we have exact interpolation. Under (ii), $Y_{rep}(\mathbf{s}_i) \sim Y_{obs}(\mathbf{s}_i)$ and $E(Y_{rep}(\mathbf{s}_i)|\text{Data}) = E(\mu(\mathbf{s}_i) + w(\mathbf{s}_i)|\text{Data}) \neq Y_{obs}(\mathbf{s}_i)$; we do not have exact interpolation.

2.5 Computer tutorials

2.5.1 EDA and spatial data visualization in R

We will illustrate our approaches using the so called WEF forest inventory data from a long-term ecological research site in western Oregon. These data consist of a census of all trees in a 10 ha stand. Diameter at breast height (DBH) and tree height (HT) have been measured for all trees in the stand. For a subset of these trees, the distance from the center of the stem to the edge of the crown was measured at each of the cardinal directions.

Rarely do we have the luxury of a complete census, but rather a subset of trees are sampled using a series of inventory plots. This sample is then used to draw inferences about parameters of interest (e.g., mean stand DBH or correlation between DBH and HT). We defer these analyses to subsequent example sessions. Here, we simply use these data to demonstrate some basics of spatial data manipulation, visualization, and exploratory analysis. We will use the following three packages:

```
> library(spBayes)
> library(classInt)
> library(RColorBrewer)
```

We begin by removing rows that have `NA` for the variables of interest, creating a vector of `DBH` and `HT`, and plotting the tree coordinates, which appear in Figure 2.12.

```
> data(WEF.dat)
> WEF.dat <- WEF.dat[!apply(WEF.dat[,c("East_m","North_m",
+        "DBH_cm","Tree_height_m","ELEV_m")], 1, function(x)any(is.na(x))),]
> DBH <- WEF.dat$DBH_cm
> HT <- WEF.dat$Tree_height_m
> coords <- as.matrix(WEF.dat[,c("East_m","North_m")])
> plot(coords, pch=1, cex=sqrt(DBH)/10, col="darkgreen",
+      xlab="Easting (m)", ylab="Northing (m)")
> leg.vals <- round(quantile(DBH),0)
> legend("topleft", pch=1, legend=leg.vals, col="darkgreen",
+        pt.cex=sqrt(leg.vals)/10, bty="n", title="DBH (cm)")
```

To further explore spatial patterns in `DBH`, it is often useful to use a color gradient or ramp to construct the pallet. These can be created using `colorRampPalette` which creates a function that interpolates over a given set of colors. Here we make a color ramp function then create a color palette with four colors, each with five shades.

```
> col.br <- colorRampPalette(c("blue", "cyan", "yellow", "red"))
> col.pal <- col.br(5)
```

We can associate the colors with intervals that are meaningful to the analysis. For instance, in a forest inventory, trees are often identified as sapling, poletimber, sawtimber, and large sawtimber classes. The following code-block produces Figure 2.13, which displays the coordinates colored according to their class.

```
> fixed <- classIntervals(DBH, n = 4, style = "fixed",
+                         fixedBreaks = c(0, 12.7, 30.48, 60, max(DBH) + 1))
> fixed.col <- findColours(fixed, col.pal)
> plot(coords, col = fixed.col, pch = 19, cex = 0.5,
+      main = "Forestry tree size classes",
+      xlab = "Easting (m)", ylab = "Northing (m)")
> legend("topleft", fill = attr(fixed.col, "palette"),
+     legend = c("sapling", "poletimber", "sawtimber", "large sawtimber"),
+     bty = "n")
```

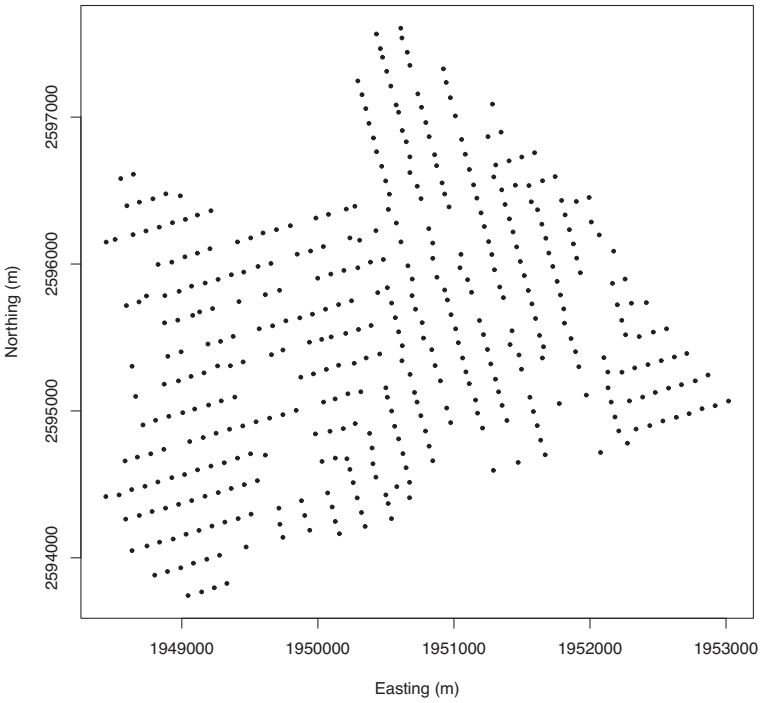


Figure 2.12 *Tree locations on the WEF.*

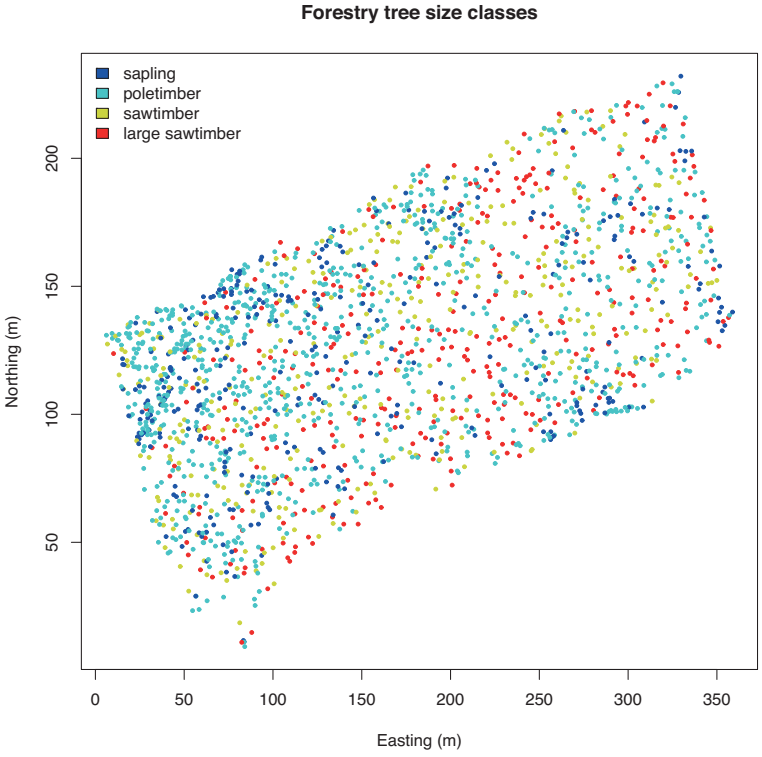


Figure 2.13 *Intervals based on previously defined tree size classes.*

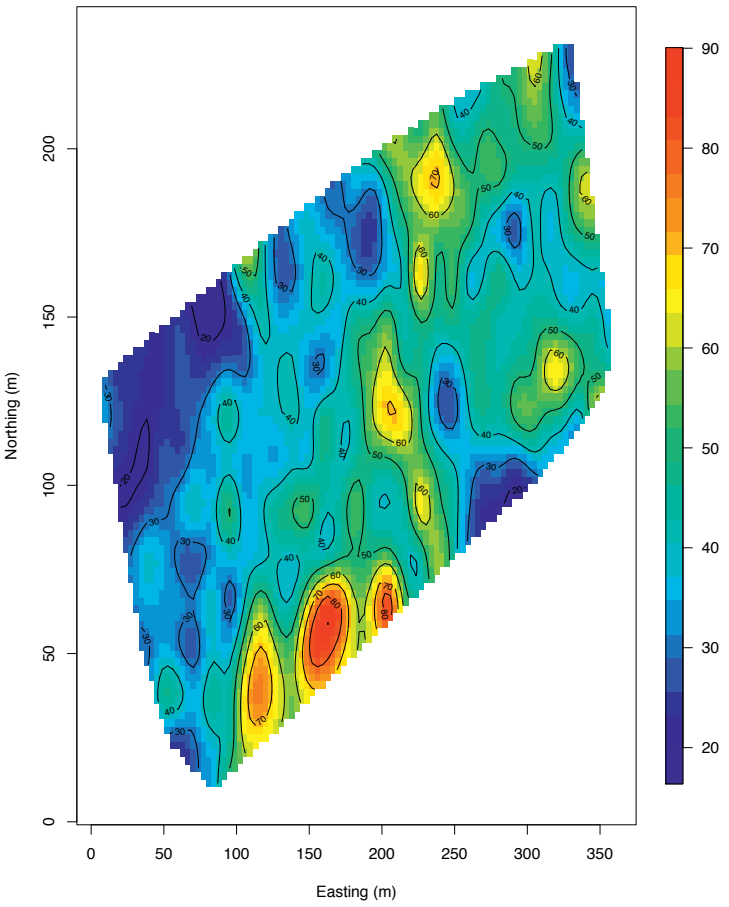


Figure 2.14 *Interpolation of DBH using multilevel B-spline.*

If the data observations are well distributed over the domain, spatial patterns can often be detected by estimating a continuous surface using an interpolation function. Several packages provide suitable interpolators, include the `akima` for linear or cubic spline interpolation and `MBA` which provides efficient interpolation of large data sets with multilevel B-splines. Functions within both packages produce grids of interpolated values which can be passed to `image` or `image.plot` to produce $\Re^2$ depictions. The code below uses `MBA` for spatial interpolation and produces Figure 2.14.

```
> library(MBA)
> library(fields) ## For using the image.plot function
> x.res <- 100
> y.res <- 100
> surf <- mba.surf(cbind(coords, DBH), no.X = x.res, no.Y = y.res,
+                  h = 5, m = 2, extend = FALSE)$xyz.est
> image.plot(surf, xaxs = "r", yaxs = "r", xlab = "Easting (m)",
+            ylab = "Northing (m)", col = col.br(25))
> contour(surf, add=T) ## (Optional) Adds contour lines to the plot
```

Alternatively, 3-D graphics functions in the `rgl` package can be used to produce $\Re^3$ depictions, as in Figure 2.15, using the code

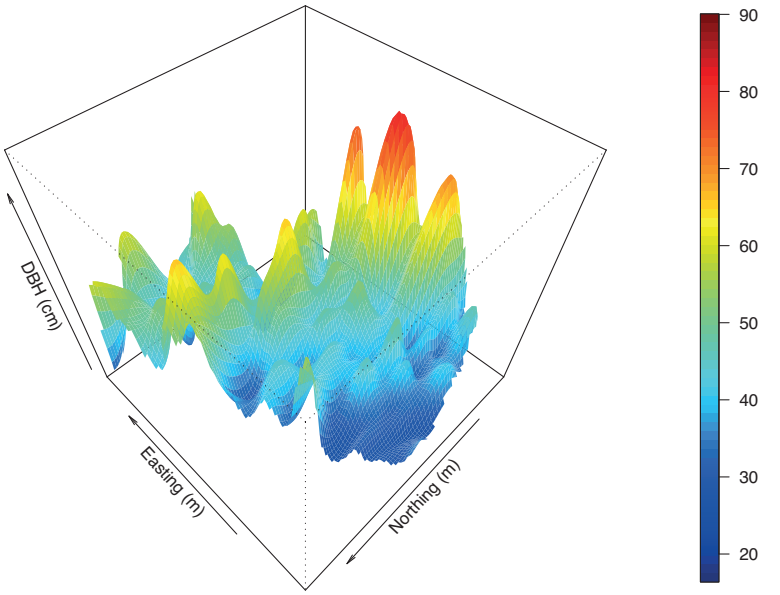


Figure 2.15 *Perspective plot of multilevel B-spline interpolation of DBH.*

```
> library(rgl)
> col <- rbind(0, cbind(matrix(drape.color(surf[[3]],
+     col = col.br(25)), x.res - 1, y.res - 1), 0))
> surface3d(surf[[1]], surf[[2]], surf[[3]], col = col)
> axes3d()
> title3d(main = "DBH", xlab = "Easting (m)", ylab = "Northing (m)",
+     zlab = "DBH (cm)")
> drape.plot(surf[[1]], surf[[2]], surf[[3]], col = col.br(150),
+     theta = 225, phi = 50, border = FALSE, add.legend = FALSE,
+     xlab = "Easting (m)", ylab = "Northing (m)",
+     zlab = "DBH (cm)")
> image.plot(zlim = range(surf[[3]], na.rm = TRUE),
+     legend.only = TRUE, horizontal = FALSE)
```

2.5.2 Variogram analysis in R

Our visual inspection of the forest inventory data suggest that there is some degree of spatial dependence in the distribution of DBH across the WEF. This encourages further exploration using a variogram analysis to quantify the range of spatial dependence and obtain an idea about the sill and the nugget.

We begin by fitting an isotropic empirical semivariogram using functions within the `geoR` package. In the code blocks provided below, we first fit an exponential variogram model to DBH, then fit a second variogram model to the residuals of a linear regression of DBH onto tree species. We first compute the inter-site distances by applying the `iDist` function to the coordinates. We then set the number of bins and use the `variog` function in `geoR` to compute an omnidirectional variogram.

```
> library(geoR)
> max.dist <- 0.25 * max(iDist(coords))
> bins <- 50
```

```
> vario.DBH <- variog(coords = coords, data = DBH,
+        uvec = (seq(0, max.dist, length = bins)))
```

Covariograms and correlograms, if desired, can be invoked using the `covariogram` and `correlogram` functions.

The computed `vario.DBH` object can then be passed to the `variofit` function in `geoR`, which uses nonlinear least squares to fit an exponential variogram model.

```
> fit.DBH <- variofit(vario.DBH, ini.cov.pars = c(600,200/-log(0.05)),
+            cov.model = "exponential", minimisation.function = "nls",
+            weights = "equal")
```

The function `variofit` can estimate the sill, the range, and the nugget parameters under a specified covariance model. A variogram object (typically an output from the `variog` function) is taken as input, together with initial values for the range and sill (in `ini.cov.pars`), and the covariance model is specified through `cov.model`. The covariance modeling options include `exponential`, `gaussian`, `spherical`, `circular`, `cubic`, `wave`, `power`, `powered.exponential`, `cauchy`, `gneiting`, `gneiting.matern`, and `pure.nugget` (no spatial covariance). Also, the initial values provided in `ini.cov.pars` do not include those for the nugget. It is concatenated with the value of the `nugget` option only if `fix.nugget=FALSE`. If the latter is `TRUE`, then the value in the `nugget` option is taken as the fixed true value.

We next fit a second variogram model to the residuals of an ordinary linear regression of DBH onto tree species. The code below produces a summary of regressing `DBH` on tree species (a categorical variable with five classes).

```
> lm.DBH <- lm(DBH ~ Species, data = WEF.dat)
> summary(lm.DBH)
Call:
lm(formula = DBH ~ Species, data = WEF.dat)

Residuals:
    Min      1Q  Median      3Q     Max
-78.423  -9.969  -3.561  10.924 118.277

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   89.423      1.303  68.629   <2e-16 ***
SpeciesGF    -51.598      4.133 -12.483   <2e-16 ***
SpeciesNF     -5.873     15.744  -0.373    0.709
SpeciesSF    -68.347      1.461 -46.784   <2e-16 ***
SpeciesWH    -48.062      1.636 -29.377   <2e-16 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 .

Residual standard error: 22.19 on 1950 degrees of freedom
Multiple R-squared: 0.5332,Adjusted R-squared: 0.5323
F-statistic: 556.9 on 4 and 1950 DF,  p-value: < 2.2e-16
```

We now collect the residuals from the above model anf fit a variogram to the residuals.

```
> DBH.resid <- resid(lm.DBH)
> vario.DBH.resid <- variog(coords = coords, data = DBH.resid,
+                           uvec = (seq(0, max.dist, length = bins)))
> fit.DBH.resid <- variofit(vario.DBH.resid,
+      ini.cov.pars = c(300, 200/-log(0.05)), cov.model = "exponential",
```

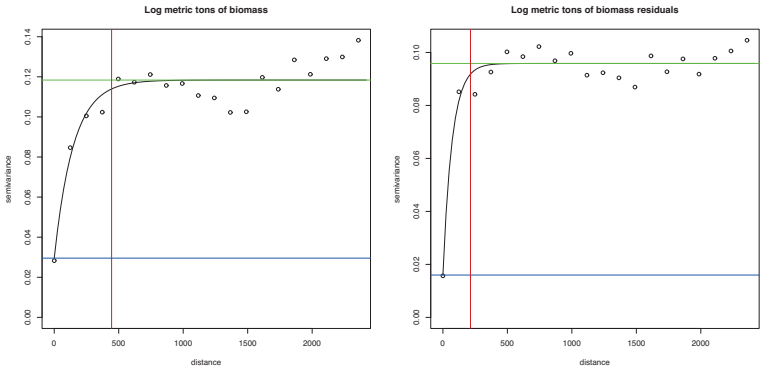


Figure 2.16 *Isotropic semivariograms for DBH and residuals of a linear regression of DBH onto tree species.*

```
+      minimisation.function = "nls", weights = "equal")
```

Finally, we plot the two variograms – one for the response and the other for the residuals – along with fitted exponential variograms.

```
> par(mfrow = c(1, 2))
> plot(vario.DBH, ylim = c(200, 1200), main = "DBH")
> lines(fit.DBH)
> abline(h = fit.DBH$nugget, col = "blue")
> abline(h = fit.DBH$cov.pars[1] + fit.DBH$nugget, col = "green")
> abline(v = -log(0.05) * fit.DBH$cov.pars[2], col = "red3")
> plot(vario.DBH.resid, ylim = c(200, 500), main = "DBH residuals")
> lines(fit.DBH.resid)
> abline(h = fit.DBH.resid$nugget, col = "blue")
> abline(h = fit.DBH.resid$cov.pars[1] + fit.DBH.resid$nugget, col = "green")
> abline(v = -log(0.05) * fit.DBH.resid$cov.pars[2], col = "red3")
```

The resulting variograms are offered in Figure 2.16. Here the upper and lower horizontal lines are the *sill* and *nugget*, respectively, and the vertical line is the effective range (i.e., that distance at which the correlation drops to 0.05).

We can also check for possible anisotropic patterns in spatial dependence. This can also be accomplished using functions in `geoR` but let us switch to a different `R` package, `gstat`, for illustrative purposes. The `gstat` package offers a `variogram` function that can be regarded as more versatile than its `geoR` counterpart in that it allows us to specify a linear regression model *within* it and conduct the variogram analysis on the residuals without having to explicitly collect the residuals. This is demonstrated in the code below. The resulting plot appears in Figure 2.17.

```
> ELEV_m <- WEF.dat$ELEV_m
> sp.dat <- as.data.frame(cbind(DBH, HT, coords, ELEV_m))
> coordinates(sp.dat) <- c("East_m", "North_m")
> vario.DBH <- variogram(DBH ~ ELEV_m, data = sp.dat,
+     cutoff = max.dist, width = 5, alpha = (0:3) *
+         45)
> fit.DBH <- fit.variogram(vario.DBH, vgm(1000, "Exp",
+     200/-log(0.05), 600))
> print(plot(vario.DBH, fit.DBH))
```

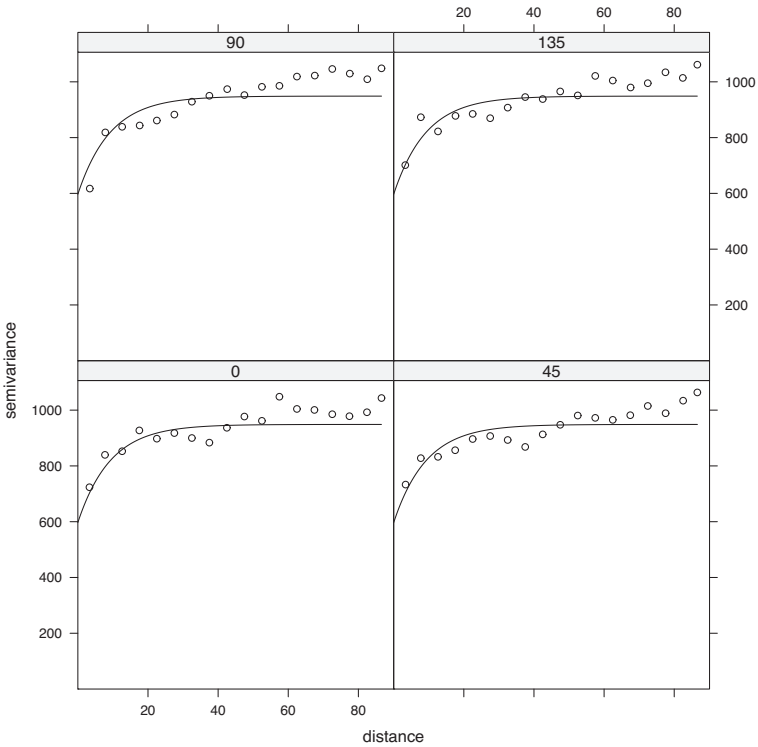


Figure 2.17 *Directional semivariograms for DBH.*

2.6 Exercises

1. Consider the time series , $Y_t = X\sin(\omega t + \theta)$ (so X is the amplitude, ω is the frequency and θ is the phase) where X is distributed with mean 0 and variance 1 independent of $\theta \sim U(-\pi, \pi)$). Show that Y_t is weakly stationary.
2. Recalling Section 2.4, show that, under a constant mean assumption, i.e., $\mu(\mathbf{s}) = \mu$, the best linear unbiased estimator (BLUE), $\hat{\mu}$ of μ under a Gaussian model is $\frac{\mathbf{1}^T\Sigma^{-1}\mathbf{Y}}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}}$. (In fact, this only requires first and second moments and not normality.) Then, show that $\boldsymbol{\ell}^T\mathbf{Y}$ is $\boldsymbol{\lambda}^T\mathbf{Y}$ with $\hat{\mu}$ replacing μ.
3. For semivariogram models #2, 4, 5, 6, 7, and 8 in Subsection 2.1.3,
   (a) identify the nugget, sill, and range (or effective range) for each;
   (b) find the covariance function $C(t)$ corresponding to each $\gamma(t)$, provided it exists.
4. Prove that for Gaussian processes, strong stationarity is equivalent to weak stationarity.
5.(a) What is the issue with regard to specifying $\widehat{c}(0)$ in the covariance function estimate (2.12)?
   (b) Show either algebraically or numerically that regardless of how $\widehat{c}(0)$ is obtained, $\widehat{\gamma}(t_k) \neq \widehat{c}(0) - \widehat{c}(t_k)$ for all t_k.
6. The scallops data can be read into `R` directly from our website by starting the program and typing

```
> myscallops <- read.table(
+ ‘‘http://www.biostat.umn.edu/~brad/data/myscallops.txt’’,
+     header=T)
```

   Carry out the steps outlined in Section 2.5.1 using `R` packages. In addition:

(a) Provide a descriptive summary of the `myscallops` data with the plots derived from the above session.

(b) Experiment with the `variog` function in `geoR` to obtain rough estimates of the nugget, sill, and range.

(c) Repeat the theoretical variogram fitting with an exponential variogram, and report your results.

7. Consider the `coalash` data frame in the `gstat` package in `R` and available from `http://www.biostat.umn.edu/~brad/data/coal.ash.txt`. This data comes from the Pittsburgh coal seam on the Robena Mine Property in Greene County, PA (Cressie, 1993, p. 32). This data frame contains 208 coal ash core samples (the variable `coal` in the data frame) collected on a grid given by x and y planar coordinates (*not* latitude and longitude).
   Carry out the following tasks in `R`:

   (a) Plot the sampled sites embedded on a map of the region. Add contour lines to the plot.

   (b) Provide a descriptive summary (histograms, stems, quantiles, means, range, etc.) of the variable `coal` in the data frame.

   (c) Plot variograms and correlograms of the response and comment on the need for spatial analysis here.

   (d) If you think that there is need for spatial analysis, arrive at your best estimates of the range, nugget, and sill.

8. Confirm expressions (2.15) and (2.16), and subsequently verify the form for $\boldsymbol{\lambda}$ given in equation (2.17).

9. Show that when using (2.15) to predict the value of the surface at one of the existing data locations $\mathbf{s}_i$, the predictor will equal the observed value at that location if and only if $\tau^2 = 0$. (That is, the usual Gaussian process is a spatial interpolator only in the "noiseless prediction" scenario.)

10. Recall that

$$\begin{aligned} \mathbf{Y} &= X\boldsymbol{\beta} + \boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} \sim N(\mathbf{0}, \Sigma) , \\ \text{and } \Sigma &= \sigma^2 H(\phi) + \tau^2 I , \text{ where } (H(\phi))_{ij} = \rho(\phi; d_{ij}) . \end{aligned}$$

    Thus the dispersion matrix of $\widehat{\boldsymbol{\beta}}$ is given as $Var(\widehat{\boldsymbol{\beta}}) = \left(X^T \Sigma^{-1} X\right)^{-1}$. Thus $\widehat{Var}(\widehat{\boldsymbol{\beta}}) = \left(X^T \widehat{\Sigma}^{-1} X\right)^{-1}$ where $\widehat{\Sigma} = \widehat{\sigma}^2 H(\widehat{\phi}) + \widehat{\tau}^2 I$ and $X = [\mathbf{1}, \texttt{long}, \texttt{lat}]$. Given the estimates of the sill, range, and nugget (from the `nls` function), it is possible to estimate the covariance matrix $\widehat{\Sigma}$, and thereby get $\widehat{Var}(\widehat{\boldsymbol{\beta}})$. Develop an `R` program to perform this exercise to obtain estimates of standard errors for $\widehat{\boldsymbol{\beta}}$ for the scallops data.

    Hint: $\hat{\tau}^2$ is the nugget; $\hat{\sigma}^2$ is the partial sill (the sill minus the nugget). Finally, the correlation matrix $H(\hat{\phi})$ can be obtained from the spherical covariance function, part of your solution to Exercise 3.